Two geographies of urban deprivation: everyday services and emergency care across European cities

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Abstract

Proximity has become the planning paradigm of European cities: the 15-minute city promises every daily need within a short walk and is advanced as a route to urban resilience. That promise concerns everyday life. Urban populations also face a growing burden of emergencies, with heat alone killing an estimated 44,000 Europeans a year and the number exposed to extreme heat projected to rise from ten million to almost 300 million a year under three degrees of warming; in an emergency what matters is whether care reaches people in time. Accessibility research measures both kinds of need as travel time, which cannot say how much a resident loses when a service lies further away. Deprivation theory, developed in humanitarian logistics to quantify the welfare lost during time without an essential good, can. We bring that theory into urban analysis and ask whether the deprivation cities impose follows the same law for everyday and for emergency needs. Measuring both on a 100 m population grid for 67 functional urban areas in 24 European countries, we find two regimes that obey different laws. Everyday deprivation falls with city size while emergency deprivation does not, and everyday inequality widens with size while emergency inequality never narrows. Emergency deprivation follows the coverage of national emergency systems rather than city size: five cities that appear average by travel time carry 2.4 to 4.7 times the typical emergency deprivation. Within cities, the neighbourhoods most deprived of everyday services are also, more often than chance, the most deprived of emergency care, and neighbourhoods with many children carry this compounded deprivation most consistently. A city that works every day is not thereby resilient.

Significance statement. Cities are planned for everyday proximity, yet urban populations increasingly face emergencies in which minutes decide survival. We show that everyday and emergency needs form two regimes of urban deprivation that obey different laws. Measuring anchored deprivation rather than travel time across 67 European city regions, larger cities reduce everyday deprivation but not emergency deprivation, everyday inequality grows with city size, and emergency deprivation follows national coverage, with five capitals harbouring emergency deserts that travel-time averages cannot detect. The same neighbourhoods, often those with many children, lose in both regimes. A city that works every day is not thereby resilient; urban resilience requires both regimes.

Keywords: urban deprivation; accessibility; 15-minute city; emergency medical services; urban resilience; spatial equity; urban scaling

1 Introduction

Three quarters of Europeans live in urban areas, a share projected to exceed 83 % by 2050 [1], and the cities they live in are increasingly planned around one idea: that every daily need should lie within a short walk. The 15-minute city has become the planning paradigm of the continent [2, 3], and it describes an everyday regime, in which residents choose when to reach a general practitioner, a pharmacy, a supermarket, a school or a park. Urban life is also lived in a second regime. Heat kills an estimated 44,000 Europeans a year today, a toll projected to triple by 2100 under current policies [4]; about ten million Europeans are exposed to extreme heat each year, a number that reaches almost 300 million, more than half the population, under three degrees of warming [5]; and the burden falls on cities, where 2.3 million climate-related temperature deaths are projected across 854 European cities by 2099 without adaptation [6]. The summer of 2026 opened with the earliest intense heatwave on record [7]. The emergency regime is exercised every day and not only in disasters: across Europe about 8 % of people who suffer a cardiac arrest outside hospital survive, the ambulance reaches them after 12 minutes on average, and each minute of delay lowers the probability of survival [8–10]. The European Union has responded by asking member states to plan for emergencies explicitly, down to 72 hours of household self-sufficiency [11]. Everyday life and emergencies place different demands on the same city. What does a resident lose when the services they need lie further away, and does the answer differ between the everyday and the emergency regime?

Urban research has one established answer, and it is a measurement rather than a theory. Accessibility, defined by Hansen as "a measurement of the spatial distribution of activities about a point" [12], is operationalised as travel time. The 15-minute-city programme [2, 13], global travel-time mapping of health care and daily services [14, 15] and the measurement of unequal access within cities [16] all count minutes, and a minute counts the same wherever it falls, for whichever service, for whichever resident. Combined with urban scaling, whose empirical regularity [17] and generative theory [18] hold that infrastructure and services improve systematically with city size, the access position implies a prediction: larger cities should serve every need better, everyday and emergency alike, because the returns to scale that concentrate facilities do not distinguish between service classes. The position cannot say how much a resident loses per minute, and it has no reason to expect two service classes to behave differently.

Welfare economics offers a second position. Deprivation theory, developed in humanitarian logistics, defines the deprivation cost of a good as the welfare lost during the time a person spends without it, and shows that this cost is estimable as a function of deprivation time [19, 20]. Two objects follow, and their difference carries this paper. For substitutable goods the loss is a *level*: bounded and S-shaped, saturating once people adapt and substitute [21]. For critical goods the loss is a *cost*: convex and unbounded, rising ever more steeply with time, with estimates for goods and specifically for emergency ambulance services [22–24]. The value-of-time literature of transport economics, built on routine trips whose delay can be rescheduled, is the wrong valuation for the second object: the clinical evidence that survival after cardiac arrest falls by one to two percentage points per minute of delay [10, 25] is a statement about mortality. Deprivation theory therefore predicts what the access position cannot: because the welfare lost follows the need, the everyday and the emergency regime need not obey the same law, and a city may serve one while failing the other. Urban resilience research reaches the same distinction from the other side. Resilience has been defined through equitable access to essential services [26, 27], the 15-minute city is advanced as a resilience strategy [2], and yet everyday vulnerability profiles have been shown to fail as predictors of crisis-time capability [28]. Social vulnerability research adds the question of who carries the burden [29]. Whether the neighbourhoods and populations that a city fails every day are also those it fails in an emergency is, in that literature, an open empirical question.

The stakes of choosing between the two positions are not academic. By the standard access statistic, the median drive time to emergency care, Bucharest is served better than the typical

European city (8.5 minutes, 0.71 times the sample median). By deprivation cost it carries 2.8 times the typical burden, because a well-served core pulls the median down while a deep periphery lies far beyond any clinical response benchmark. Which number a planner sees decides whether Bucharest has an emergency problem at all. In this paper we measure which of the two positions describes urban Europe.

We do so for 67 functional urban areas (FUAs) in 24 European countries, from 101,050 to 12.9 million inhabitants, drawn by a stratified design over four macro-regions and four size strata, on a common 100 m population grid. Europe is the setting for three reasons. First, comparability: Europe has a harmonised functional city definition [30], a harmonised 1 km census grid, near-complete street and facility mapping, and clinical emergency-response benchmarks shared across national systems [31–33]; global travel-time mapping exists [14, 15], but it cannot carry a welfare calibration or a census-anchored vulnerability analysis. Second, maturity: large cities lose their growth advantage as urban systems mature [34], and Europe is the mature end of that life cycle, so its cross-sectional size gradient reads what city size delivers once growth no longer favours the large city. Third, obligation: access to services of general economic interest and to essential services is a right in the European legal order [35, 36], emergency care is organised nationally under shared response targets, and the Preparedness Union Strategy now asks member states to plan for the emergency regime explicitly [11]. The sample therefore covers one polity with one set of obligations and 24 national emergency systems. Following the scaling tradition we estimate log-log elasticities of city outcomes on population [17] under the harmonised city definition [34]; every size gradient is cross-sectional, and any phrase such as "as cities grow" means the space-for-time reading of that gradient, never a longitudinal observation.

Three findings follow. First, deprivation is a different object from access: it explains the cross-city structure better than travel time, agrees with travel time wherever results are rank-based, and reveals what travel-time averages hide, five emergency deserts in European capitals. Second, urban growth delivers on one clock only: everyday deprivation falls with city size and emergency deprivation does not, everyday inequality widens with size and emergency inequality never narrows, and emergency deprivation follows the coverage of national emergency systems rather than city size. Third, the two deprivations compound: in 66 of 67 cities the neighbourhoods most deprived of everyday services are also, more often than chance, the most deprived of emergency care, most strongly in Central-Eastern Europe, and neighbourhoods with many children carry the compounded burden most consistently.

2 Results

Two surfaces summarise each city; their construction is in Materials and Methods, and Fig. 1 shows the two deprivation functions. The *everyday deprivation level* of a grid cell, between 0 and 1, expresses how far the burden of reaching five everyday service categories on foot has progressed toward saturation: near 0, all services are comfortably walkable; near 1, the cell is effectively without walkable everyday services and additional minutes no longer add burden. The *emergency deprivation cost* of a cell is an open-ended positive number expressing the welfare lost through its drive time to the nearest emergency facility, in multiples of the loss at the 15-minute emergency-medical-service (EMS) benchmark: a cell at 2.0 carries twice the deprivation of one whose ambulance arrives at the benchmark, and the cost keeps escalating beyond it. "Cost" here is welfare lost, never money. Deprivation throughout this paper is the welfare loss imposed by poor physical access to a service. It is distinct from the socioeconomic deprivation of income, employment or housing that composite deprivation indices measure, which enters the analysis only as an explanatory variable, through the vulnerability strata of Section 2.4. Both curves are anchored on external standards rather than fitted to the data: the 15-minute-city threshold for the everyday level, and the EMS response benchmarks for the emergency cost.

The level and the cost are different objects because the needs are, and deprivation theory

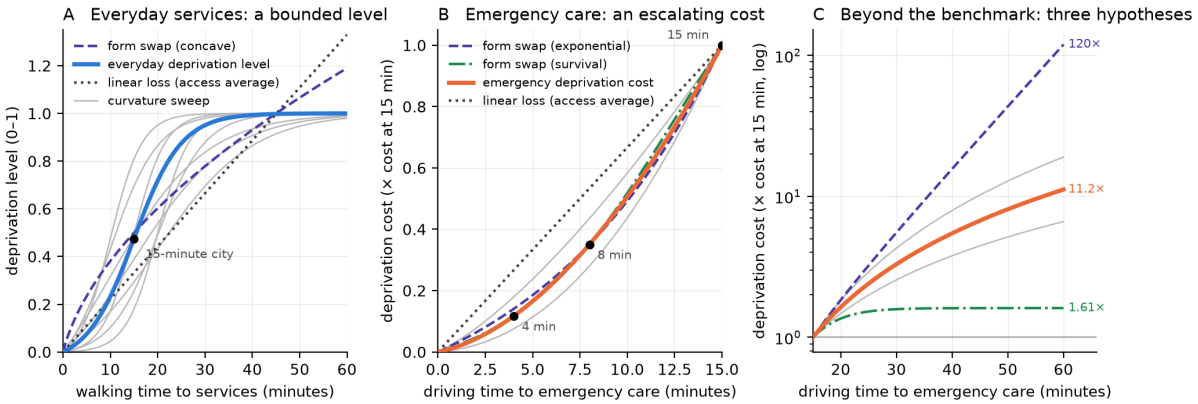


Figure 1: The two deprivation functions. (A) Everyday deprivation level against walking time: bounded, saturating, with its inflection at the 15-minute anchor. (B) Emergency deprivation cost against driving time inside the benchmark window, in multiples of the cost at the 15-minute EMS target; all forms coincide at the clinical anchors (4, 8 and 15 minutes) by construction. (C) Beyond the 15-minute target no benchmark exists and three literature-grounded escalation hypotheses diverge (log scale). Grey lines are the curvature sweep and dashed lines the form swaps used in the robustness analysis (Materials and Methods); the dotted line is the linear loss that any travel-time average implicitly applies. Panels are on different scales and are never compared in magnitude.

Table 1: Sample composition and median city indicators by macro-region. 67 functional urban areas (Eurostat/OECD definition: city plus commuting zone), stratified draw over 4 macro-regions × 4 size strata (10^5–$2.5 \cdot 10^5$, $2.5 \cdot 10^5$–10^6, 10^6–$5 \cdot 10^6$, $> 5 \cdot 10^6$ inhabitants; 15/18/30/4 cities), 113.9 million residents in total. $\bar{D}_{\text{ev}}$: population-weighted mean everyday deprivation level (0–1); $\bar{C}_{\text{em}}$: mean emergency deprivation cost (multiples of g(15); not comparable to $\bar{D}_{\text{ev}}$ in magnitude); G: within-city population-weighted Gini; ρ: Spearman rank coupling of the two surfaces; HH: compounding population share at the median split. Source: `cities_descriptives.csv`.

Region	Cities	Countries	Pop. (M)	$\bar{D}_{\text{ev}}$	$\bar{C}_{\text{em}}$	G_{ev}	G_{em}	ρ	HH
North	13	4	11.3	0.41	1.81	0.39	0.47	0.47	0.34
West	16	5	37.1	0.23	0.74	0.52	0.41	0.36	0.32
South	18	4	39.8	0.29	1.07	0.42	0.42	0.38	0.32
CEE	20	11	25.6	0.30	1.48	0.51	0.54	0.55	0.36
All	67	24	113.9	0.29	1.14	0.49	0.44	0.42	0.33

says why. The welfare lost during time without a good depends on whether the good can be substituted or deferred [19]. Everyday services can: a further pharmacy, a later trip or a delivery absorbs the loss, so the marginal minute loses value once coping sets in and the loss is bounded. Emergency care cannot: survival worsens with every minute [10, 25], so the marginal minute gains value and the loss is unbounded. A single loss function for both would either declare a 60-minute ambulance no worse than a 30-minute one or declare a distant supermarket a catastrophe. Relative to the calibrated curves, the linear loss of a travel-time average over-prices minutes where they are welfare-inert (the saturated everyday range) and under-prices them where they are clinically decisive (around the emergency benchmarks). The asymmetry imposes one discipline used throughout: only unit-invariant statistics cross regimes (elasticities, Ginis, rank-based measures), and raw magnitudes never do.

Table 1 summarises the sample; Table S11 in the SI lists every city. Because cities nest in 24 countries, every scaling claim below cites a wild cluster bootstrap p-value on country clusters and every regional contrast a country-permutation p-value; the inference conventions are in Materials and Methods.

Table 2: The five emergency-desert capitals: access averages against deprivation. Median drive time to the nearest emergency facility and its ratio to the 67-city sample median (12 min) against mean emergency deprivation cost (multiples of g(15)) and its ratio to the sample value (1.14). Source: `desert_access_contrast.csv`.

City	Country	Median time (min)	Time ratio	Mean cost	Cost ratio
Vilnius	LT	14.0	1.17	5.39	4.72
Ljubljana	SI	23.0	1.92	4.96	4.34
Rīga	LV	13.5	1.13	3.37	2.95
Bucureşti	RO	8.5	0.71	3.21	2.81
Tallinn	EE	13.0	1.08	2.80	2.45

2.1 Deprivation is a different object from access

The first result is that the deprivation layer changes what is seen. Every headline regression of this paper is estimated twice, once on the deprivation surfaces and once on the deprivation-free, minutes-based indicators the accessibility literature would use [15, 16], and the comparison has three parts (Fig. 2; exact values in Table S9).

Deprivation is the better-behaved object. City size explains 44 % of the cross-city variance in mean everyday deprivation against 16 % for the median walk minute, its access counterpart (Fig. 2A). The reason is the saturating level function: whether a periphery cell sits 50 or 70 minutes from a pharmacy moves a travel-time average and moves no household's daily routine, and the level function discounts that variation. Anchored deprivation acts as a welfare-weighted filter on travel time, and the filtered outcome is the one that scales cleanly. The filtering costs nothing in ordering: the deprivation-based and time-based Gini families rank cities almost identically (Pearson $r = 0.965$ everyday, 0.976 emergency), which is why every rank-based result in this paper can be called parameterisation-free, and why the level and inequality results are where the calibration carries information that travel time cannot.

Deprivation reveals what travel-time averages hide. Bucharest's median emergency drive time is 8.5 minutes, 0.71 times the sample median, so by the standard access statistic the city is better served than typical urban Europe. Its mean emergency deprivation cost is 2.81 times the sample value (Fig. 2B; Table 2). Four other capitals, Vilnius, Ljubljana, Rīga and Tallinn, show the same anatomy: median-time ratios of 0.71–1.92 coexist with deprivation ratios of 2.45–4.72, because a well-served core pulls the median minute down while a deep periphery, where drive times run far beyond the 15-minute benchmark, carries convex costs that no average over uniform minutes can register. These five cities are the emergency deserts of Section 2.3, and only a loss function that prices the tail the way the clinical benchmarks say it should reveals them. Even inside the access paradigm the signal sits in the tail: the only minutes-based emergency statistic to reach conventional significance in the size regressions is the 90th-percentile drive time (-0.059, $p = 0.017$), the most tail-sensitive access statistic in the set.

Deprivation agrees with access wherever the result is about ranks, and that agreement protects the findings below from the suspicion of being calibration artefacts. The everyday size gradient is present in plain minutes (median walk-time elasticity -0.237, $p = 3.8 \times 10^{-5}$; mean walk time -0.178), and the absence of an emergency gradient is present in minutes too (elasticities -0.043 to -0.059 across median, mean and 90th-percentile drive time). No headline of Sections 2.2–2.3 depends on the deprivation functions; the functions restate in welfare terms a structure the network geometry already carries, and add the levels, the inequality magnitudes and the tails.

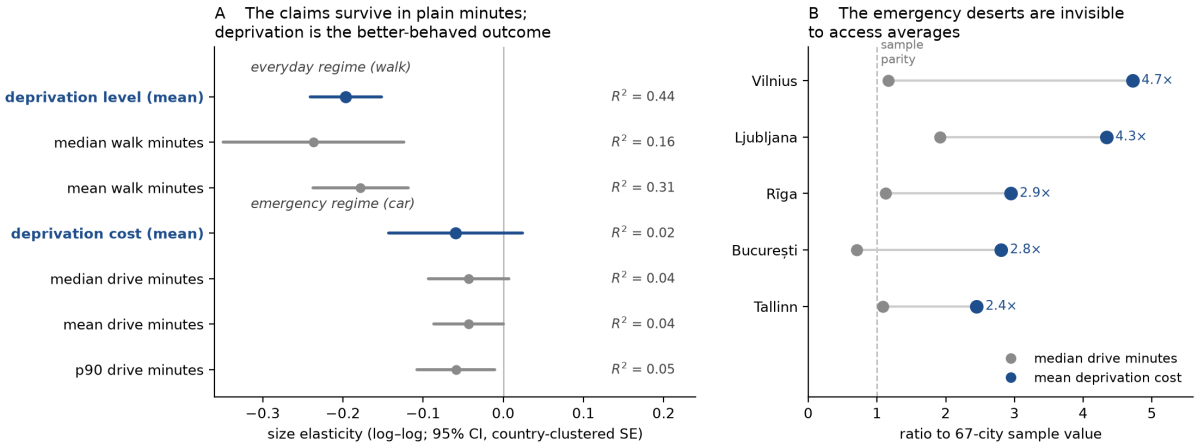


Figure 2: Deprivation against access. (A) Size elasticities of deprivation-based (blue) and minutes-based (grey) outcomes per regime (points: estimates; bars: 95 % CIs on country-clustered SEs; annotations: R^2). The everyday gradient and the absence of an emergency gradient are present in both families; the deprivation outcomes are the better-behaved ones. (B) The five emergency-desert capitals: ratio to the 67-city sample value under the access average (median drive minutes, grey) and under mean emergency deprivation cost (blue). All five sit near or below parity by access and at 2.45–4.72 times the sample by deprivation. Sources: `deprivation_vs_access.csv`, `desert_access_contrast.csv`.

2.2 Larger cities have lower everyday deprivation, but not lower emergency deprivation

If the returns to scale behind urban service provision were general, as scaling theory holds for infrastructure [17, 18], both deprivations should fall with city size. The everyday regime behaves as that account predicts (Fig. 3A). Mean everyday deprivation falls systematically with population: the log-log elasticity is −0.196 (country-clustered SE 0.023, $p_{\text{wild}} = 0.0002$), so mean everyday deprivation falls by roughly 19 % for each e-fold (≈ 2.7×) of population, and city size alone explains 44 % of the cross-city variance. The mechanism is visible beneath the deprivation surfaces: the median FUA routes its population to 93 mapped general practitioners, 228 pharmacies, 325 supermarkets, 409 schools and 636 green spaces, the population-weighted median walk to the nearest facility of each everyday category is 4–8 minutes, and larger cities carry denser facility networks. The estimate is stable under country fixed effects (−0.17), under leave-one-out city deletion (envelope −0.204 to −0.185, with Athina the most influential city), under outlier-robust estimators (Huber −0.196, median regression −0.223) and under exclusion of the five emergency-desert capitals (−0.203), so no single city, country or estimator choice carries it.

The emergency regime breaks the pattern. Mean emergency deprivation cost shows no detectable size gradient: the elasticity is −0.059 (clustered $p = 0.16$, $p_{\text{wild}} = 0.21$), and city size explains 2 % of the cross-city variance. Non-significance alone would be weak evidence, so the null is bounded: two one-sided equivalence tests place the emergency elasticity within ±0.13 [37]. The bound licenses a precise statement and no more: any emergency size gradient, if present, is at most two-thirds the everyday one in magnitude; equivalence at the stricter ±0.10 bound is not shown, so the claim is “no detectable scaling, bounded within ±0.13”, never “proven flat”. The descriptive layer shows why: the median FUA has 6 emergency departments and 6 ambulance stations for its whole territory, the median population-weighted drive to either is 12 minutes, and the facility count of a national emergency system does not multiply with city population the way pharmacies do. The two slopes also differ from each other, tested within cities rather than by comparing two separate regressions: a paired regime test (regime × log-population interaction, country-clustered) puts the everyday slope 0.137 below the emergency slope ($p_{\text{wild}} = 0.002$). Agglomeration benefits are therefore regime-specific, and no single exponent describes what

city size does to service deprivation. Two qualifications belong here. The minimum detectable emergency elasticity at 80 % power is 0.12, so the design could have detected a gradient of the everyday magnitude. And the absence of a linear gradient does not make size irrelevant to emergency deprivation: a quadratic term is significant (wild $p = 0.001$; $p = 0.010$ with coverage-grade dummies), an inverted U in which the middle size strata carry the highest emergency deprivation cost and the four largest FUAs the lowest (SI S4). Because those four cities are the whole top stratum, we report the shape and do not headline it.

Within-city inequality moves in the opposite direction (Fig. 3B). The Gini of everyday deprivation rises with population: elasticity +0.062 ($p_{\text{wild}} = 0.0012$). The emergency Gini is size-flat at the baseline parameterisation (+0.004, $p_{\text{wild}} = 0.84$), and the paired slope difference of −0.058 is significant over all 67 cities ($p_{\text{wild}} = 0.029$). Size therefore buys lower mean everyday deprivation at the price of a wider spread around that mean, while emergency inequality is set elsewhere. The everyday half of the claim is defended by a specification curve [38] rather than by any single calibration (Fig. SS2): re-estimating the slope under all 14 deprivation parameterisations keeps it positive throughout (+0.026 to +0.075) and significant in 13 of 14; the exception is the concave Box-Cox everyday form swap (+0.026, $p = 0.083$). The emergency half is conditional, and the conditionality is a finding. The EMS benchmarks define the cost scale up to the 15-minute cut-off, where all three literature-grounded escalation forms coincide; beyond the cut-off no benchmark exists, and there the forms diverge (Fig. 1C). Under the bounded survival-based curve the emergency Gini is size-flat (+0.001, $p = 0.97$); under polynomial Box-Cox escalation, the published baseline, it is size-flat across the whole curvature sweep (−0.006 to +0.014, none significant), in agreement with the deprivation-free travel-time Gini, itself size-flat (−0.015, $p = 0.42$); under exponential escalation it rises with size (+0.098, $p = 0.0003$), because exponential pricing turns the Gini into an index of extreme-tail depth (it saturates at a median of 0.96, decouples from the time distribution's inequality, $\rho = 0.04$, and tracks the beyond-45-minute population share, $\rho = 0.45$), and how many minutes past the cut-off the worst-served residents sit does grow with size even though the tail's population share does not (beyond-30-minute share elasticity −0.003, $p = 0.99$). Two statements hold under every parameterisation: everyday inequality rises with size, and emergency inequality never falls with size. Whether emergency inequality is flat or rising is decided by the beyond-benchmark escalation form, which no clinical benchmark pins down; it is flat under two of the three hypotheses and rising under the third.

2.3 Emergency deprivation follows national emergency-system coverage, not city size

City size explains 2 % of the variance in mean emergency deprivation cost. Most of the remaining variance lies between countries: in a nested decomposition of the 67 city values, 25 % of the variance sits between the four macro-regions, a further 54 % between countries within regions, and only 22 % between cities of the same country (Table S2). The same holds for the inequality indicators: the country level accounts for 70 % of the variance in the emergency Gini, 75 % in the everyday Gini and 76 % in the divergence gap, against 40 % for the coupling and compounding measures. We first identify the structure in the city vectors themselves, then ask what geography it has, and then test it at the level at which it varies, the country.

A coverage ordering, not a typology. The 67 cities do not fall into types. k-means clustering on the full standardised city vector separates 62 cities from 5 (silhouette 0.65, bootstrap ARI 0.87 [39, 40], Gaussian-null $p = 0.001$), and the isolated group is an outlier group rather than a type: its agreement with the region partition is ARI = 0.001. Those five are the emergency-desert capitals of Section 2.1, separated by their beyond-threshold emergency travel-time tails. Ward clustering selects the same axis and cuts it more conservatively, isolating a nested subset of the two most extreme capitals (65/2). The nesting is evidence for a gradient: the two

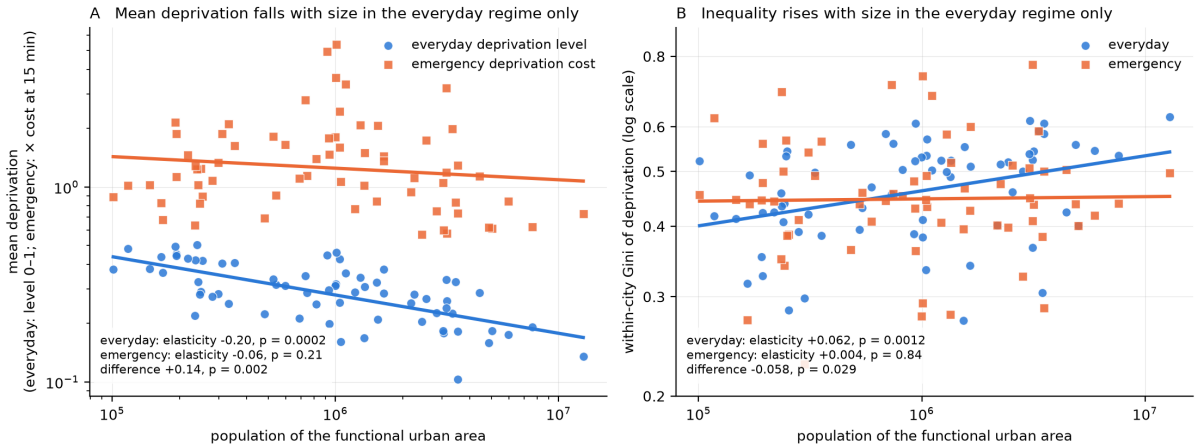


Figure 3: What city size does to deprivation, per regime. (A) Population-weighted mean deprivation against FUA population (log-log) for the 67 cities; the two series are in different units (everyday: level 0–1; emergency: cost in multiples of g(15)), so only the slopes compare, never the levels. Everyday: elasticity -0.196, wild bootstrap $p = 0.0002$, $R^2 = 0.44$; emergency: -0.059, $p = 0.21$, bounded within ± 0.13 by equivalence testing; paired slope difference $+0.137$, $p = 0.002$. (B) Within-city Gini of each surface against population (log-log): everyday elasticity $+0.062$ ($p = 0.0012$), emergency $+0.004$ ($p = 0.84$), paired difference -0.058 ($p = 0.029$). Fitted lines are the OLS fits the elasticities come from; inference is country-clustered. Gradients are cross-sectional (space-for-time).

algorithms disagree about where to cut a severity ordering and agree on the ordering itself. Re-clustering with the outliers removed finds one further split along the same axis at lower intensity, a 48/14 cut isolating partial-desert cities with real emergency tails (among them Oslo, Stockholm, Warszawa, Sofia, Zagreb, Palermo and Luxembourg; silhouette 0.38, bootstrap ARI 0.74, null $p = 0.001$, ARI vs region 0.07), inside which Ward again nests (56/6). A feature-set robustness check that re-runs the whole pipeline with the 8- and 15-minute benchmark shares added reproduces the main partition exactly (ARI 1.00) and the peeled cut at ARI 0.97, while flagging Brăila and Łomża, the two smallest CEE cities, as sitting on the covered/partial boundary. The citable structure is therefore a one-dimensional *coverage ordering* of the emergency periphery, covered (48) → partial desert (14) → desert (5), with a soft boundary between covered and partial and a robust ordering. We call these coverage grades and never city types. A city's grade expresses how much of its population lives beyond the emergency response benchmark: covered cities have a thin tail, partial deserts a real one, deserts a periphery in which drive times run far past the 15-minute target.

The coverage grade, and no other variable we test, sets the emergency level. In a single country-clustered model with ln population centred at the sample mean, the grade dummies shift mean emergency deprivation cost overwhelmingly (Wald $p \approx 1.3 \times 10^{-15}$); the median mean cost rises 1.02 → 1.87 → 3.37 multiples of g(15) across covered → partial → desert (Fig. 4). The slope-by-grade interaction is $p = 0.010$ under the current 48/14/5 grades but was $p = 0.17$ under the pre-refresh 50/12/5 grades; a two-city boundary change flips it, so we treat the interaction as boundary-sensitive and do not headline any grade-specific size gradient. Consistent with this, the divergence-gap measures show at most a weak size gradient (gap slope -0.066 per $\log_{10}$ population, $p = 0.012$, $R^2 = 0.06$; Fig. SS4), so size is a poor predictor of where the two regimes part ways.

The geography of the ordering. The coverage ordering has a geography (Fig. 5). All five deserts are capitals of Central-Eastern European (CEE) countries. The partial-desert grade is a Nordic and CEE phenomenon: 7 of the 13 Nordic cities and 5 of the 20 CEE cities carry a real emergency tail, against 1 of 16 in the West (Luxembourg) and 1 of 18 in the South

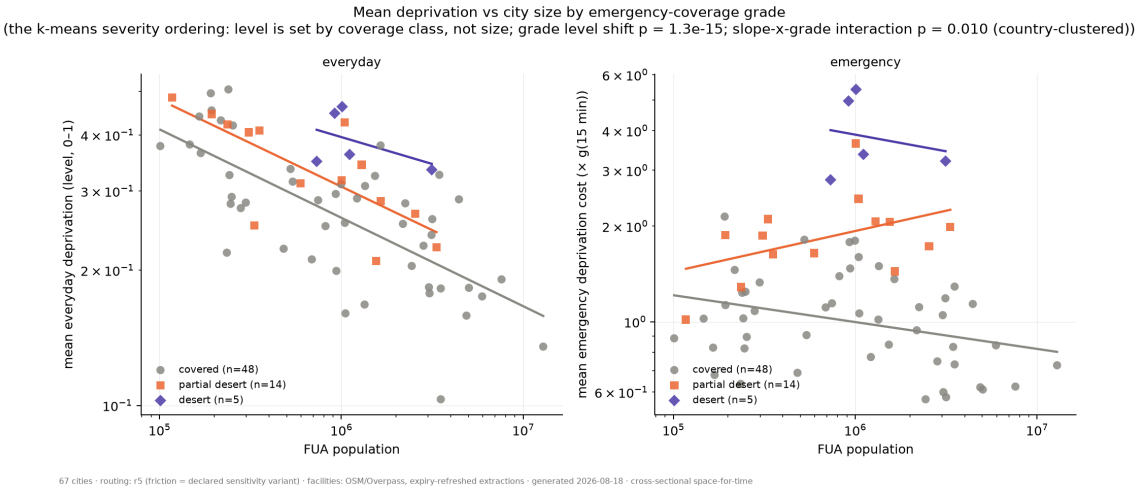


Figure 4: Mean deprivation against city size by emergency-coverage grade, per regime (left: everyday level; right: emergency cost, multiples of g(15)), grades read from the two clustering passes (covered 48 / partial desert 14 / desert 5). On the emergency side the grade shifts the level (country-clustered Wald $p \approx 1.3 \times 10^{-15}$; median levels 1.02 / 1.87 / 3.37), while the slope-by-grade interaction is boundary-sensitive and not headlined. Within-grade fits are plain OLS over few cities, shown for orientation only.

(Palermo) (Table S5). West and South are almost entirely covered. The two partitions, grade and region, are therefore different (ARI 0.001 and 0.07) and related: the emergency tail that defines the grades is a feature of the sparsely settled peripheries of Nordic FUAs and of the national emergency systems of Central-Eastern Europe. In the inequality plane (Fig. 6) the five deserts are the five highest points on the emergency-Gini axis and the partial deserts fill the band below them, so the emergency-side inequality of a city is, to a first approximation, its coverage grade. Across cities the two Ginis are uncorrelated ($r = 0.12$, $p = 0.32$; Table S8): how unequally a city distributes everyday deprivation says nothing about how unequally it distributes emergency deprivation, which is the cross-city face of the two-regime claim and points at two different policy levers.

Regional contrasts, tested at the country level. Because macro-region is a property of countries, whole countries are permuted across regions in every regional test (Fig. 7). The emergency Gini differs by macro-region (Cohen $f = 0.74$, $p_{\text{perm}} = 0.0024$), as do the divergence gap ($p_{\text{perm}} = 0.010$), the everyday–emergency coupling ρ ($p_{\text{perm}} = 0.0015$) and both compounding measures ($p_{\text{perm}} = 0.0008$ for the HH share, $p_{\text{perm}} = 0.0023$ for the continuous intensity). Central-Eastern Europe compounds: the region pairs the highest emergency inequality (median emergency Gini 0.54) with the strongest coupling between the two surfaces (median $\rho = 0.55$), so the same populations tend to carry both deprivations. The West's inequality problem is everyday-side (median divergence gap -0.12); the North's is emergency-side ($+0.09$). These contrasts are not an artefact of the five deserts: re-testing without them leaves every CEE contrast in place (coupling $p_{\text{perm}} = 0.003$, emergency Gini $p_{\text{perm}} = 0.006$, divergence gap $p_{\text{perm}} = 0.035$, HH share $p_{\text{perm}} = 0.0016$, compounding intensity $p_{\text{perm}} = 0.0023$; Table S4). The regions also differ in how dispersed their cities are: the spread of the emergency Gini and of the divergence gap is about twice as wide in Central-Eastern Europe as elsewhere (standard deviation 0.13 against 0.06–0.08 for the emergency Gini), and the difference in dispersion survives country permutation ($p_{\text{perm}} = 0.028$ and $p_{\text{perm}} = 0.010$; Table S3), while the dispersion of the coupling and of the everyday Gini does not differ regionally.

The everyday Gini's apparent regional contrast does not survive country-level testing ($p_{\text{perm}} = 0.27$ despite a city-level ANOVA $p = 0.0007$), and Fig. 7 shows why. The Southern cities split into two groups with no overlap: the seven Spanish cities and Athina have everyday Ginis of

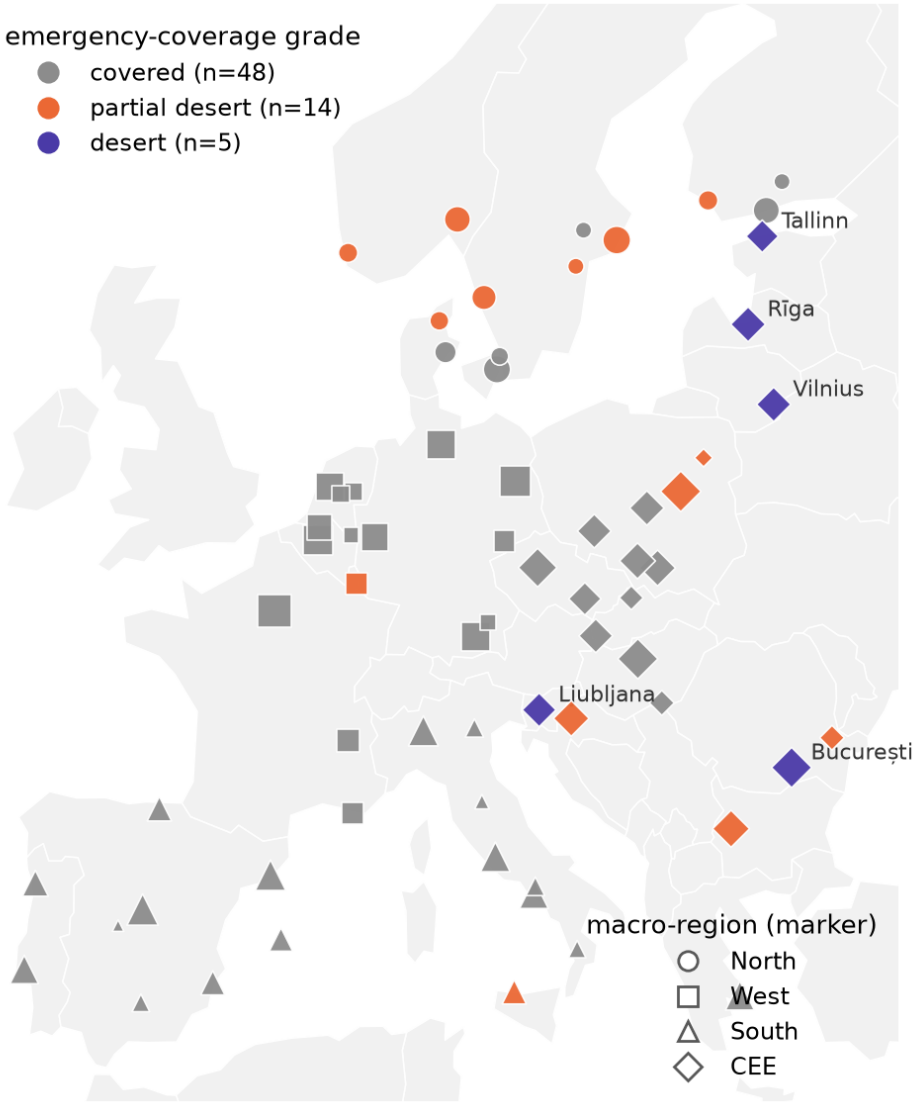


Figure 5: The 67 functional urban areas by emergency-coverage grade (fill) and macro-region (marker), sized by population. The five deserts are Central-Eastern European capitals; the partial-desert grade concentrates in the Nordic countries and Central-Eastern Europe; West and South are almost entirely covered. Point locations are city centres.

0.50–0.61, the ten Italian and Portuguese cities 0.27–0.42 (Table S6). Half of the variance in the everyday Gini lies between countries within regions, more than for any other indicator, so the "regional" contrast is a country contrast. It coincides with a known mapping artefact: primary care in Portugal and Italy sits in health centres that OpenStreetMap tags differently from individual practices, so mapped general-practitioner density there is a fifth of the Western median (Section 3.5), which overstates everyday deprivation levels and compresses their within-city spread. The paper therefore makes no regional everyday-Gini claim and no cross-country comparison of everyday levels; the everyday size gradients are unaffected, because mapped GP density is uncorrelated with city size.

2.4 Compounding is the norm, and children carry it

The co-location typology measures how often the same residents lose in both regimes. It cuts both percentile surfaces at their population-weighted medians, which fixes the marginals at one half each; the whole 2×2 class table then has a single free number, the high-high (HH)

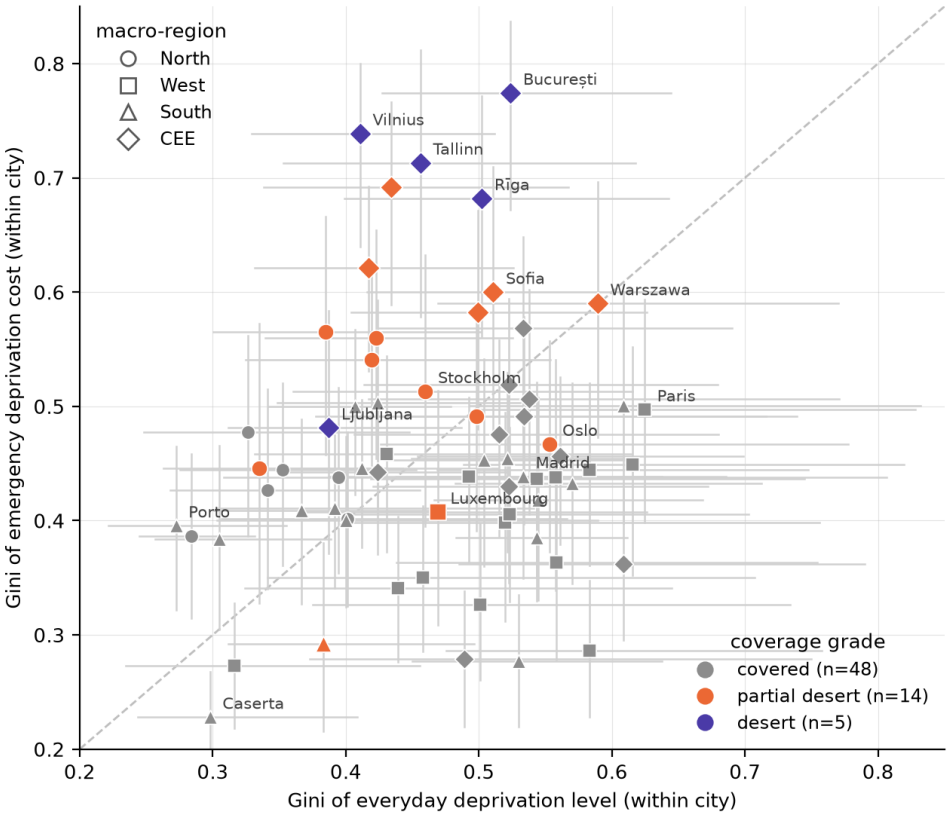


Figure 6: All 67 cities in the inequality plane. Horizontal axis: within-city Gini of the everyday deprivation level; vertical axis: within-city Gini of the emergency deprivation cost. Fill marks the coverage grade, marker shape the macro-region, and whiskers span the min–max envelope of each Gini over the deprivation-curvature sensitivity grid (Section 3.5; SI S2), so each city is placed at the precision the calibration supports. The deserts and partial deserts occupy the top of the emergency axis; the everyday axis is ordered by country (Fig. 7).

compounding share, and statistical independence of the two surfaces corresponds to HH = 25 %. Everything above 25 % is excess co-location. The observed HH share averages 33 % and exceeds independence in 66 of 67 cities; the rank coupling ρ between the two surfaces spans -0.02 to 0.74 (Fig. SS3). Compounding is not an exotic overlap of two rare conditions; across urban Europe it is the default relationship between the two deprivations.

Thresholded shares invite the question of the threshold, so the robustness harness prices it: the "how high is high" percentile cut moves the HH share by 29.8 percentage points at the median city, against 1.4 under the whole deprivation-curvature grid and 0.9 under the functional-form swaps, a roughly 21-fold dominance of the analyst's threshold over the welfare calibration. Two consequences follow. First, the headline compounding measure is the threshold-free continuous intensity, the population-weighted mean depth to which the typical resident sits inside *both* percentile distributions, with fixed anchors (1/3 under independence, 1/2 under perfect coupling, 1/4 under perfect divergence); it shows the same regional structure as the thresholded share ($p_{\text{perm}} = 0.0023$, strongest in Central-Eastern Europe). Second, every thresholded share in this paper travels with its threshold sweep. The spatial anatomy is consistent across contrasting cities (Fig. 8): compounding concentrates in peripheries where the saturated everyday level meets the escalating emergency cost, while dense cores hold most population in the low-low class. The maps are read under two rules stated once: legends carry population-weighted shares while the picture is area-weighted (a sparse periphery dominates visually at a modest population share), and no single cell's class is a finding (roughly 24 % of population flips class between routing engines); the per-city maps for all 67 cities are in the SI gallery.

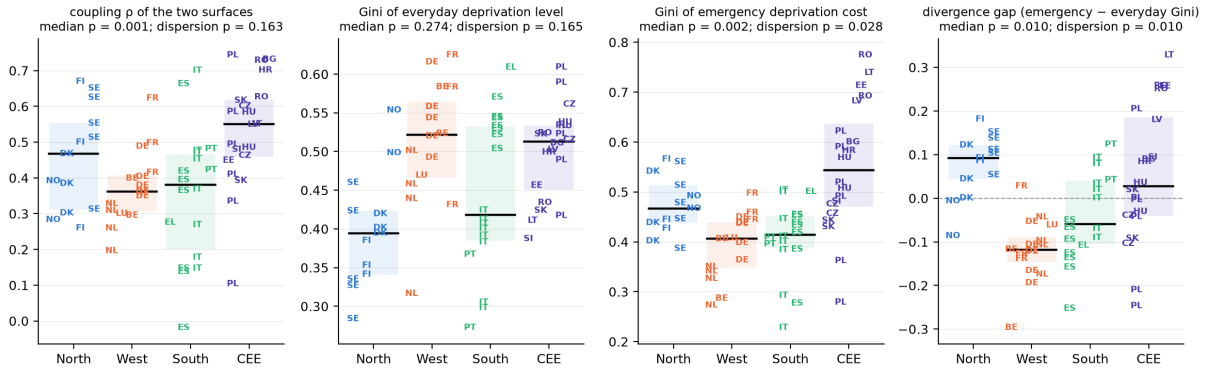


Figure 7: Regional structure of the divergence, each city drawn as its country code (bar = regional median; band = interquartile range) for the coupling ρ, the everyday Gini, the emergency Gini and the divergence gap. Each panel title gives the country-permutation p-value of the regional contrast in medians (`inference_regional.csv`) and of the contrast in dispersion (`regional_dispersion.csv`). The everyday-Gini panel shows the Spanish and Greek cities above the Italian and Portuguese ones with no overlap, which is why that contrast is a country contrast and is not claimed regionally. Region is a property of countries, so whole countries are permuted across regions; city-level p-values would be anti-conservative.

The vulnerability analysis asks who lives in the compounded areas. Following the social-vulnerability tradition [29], vulnerability enters on two strictly separated levels: EU-harmonised census age strata, comparable across all 67 cities and pooled; and national fine-grid strata (rent, tenure, finer ages), which are per-city depth and never pooled. Each stratum is compared against the cells where its own census variable is published (the covered-cell reference), because data coverage has its own geography. At the harmonised level the result is one-sided: cells in the top quartile of child share carry more everyday deprivation than their comparison cells in 88 % of cities (median ratio 1.27) and more compounding in 88 % (median HH-share gap +0.13), strongest in Central-Eastern Europe (median ratio 1.44) and the North (1.30) (Fig. 9). The elderly show no such pattern at the European level: median ratio 0.96, above parity in only 37 % of cities, below parity across most of Central-Eastern Europe. That null is itself a result, and it is scale-specific: the national-data tier resolves sharper per-city patterns (in Hamburg, elderly 1.37, children 1.52, low-rent 1.61 against covered references), but those use different definitions and stay per-city. The European claim is about children. Whatever sorting mechanism places families with children in the compounded periphery operates consistently across nearly the whole sample, in a group whose daily radius is short and whose emergency risk is real.

3 Discussion

3.1 Deprivation as the welfare measure of urban proximity

The paper's first contribution is to bring deprivation theory from humanitarian logistics [19, 20] into the analysis of a functioning city, and the results show what the transfer buys. Where results are rank-based, travel time and deprivation agree almost perfectly ($r \geq 0.965$ between the Gini families), which certifies the accessibility literature's rank-level findings [15, 16] at continental scale. Where results concern levels, inequality magnitudes or tails, travel time was never a welfare measure, and the divergence is not noise: the desert capitals sit at 0.71–1.92 times the sample by access and at 2.45–4.72 times it by deprivation (Table 2), and the choice of measure decides whether an emergency-side size story exists at all. The same limitation is conceded from inside the resilience literature, where the latest building-level frameworks measure access as network distance to the nearest facility and note that distance captures one dimension of

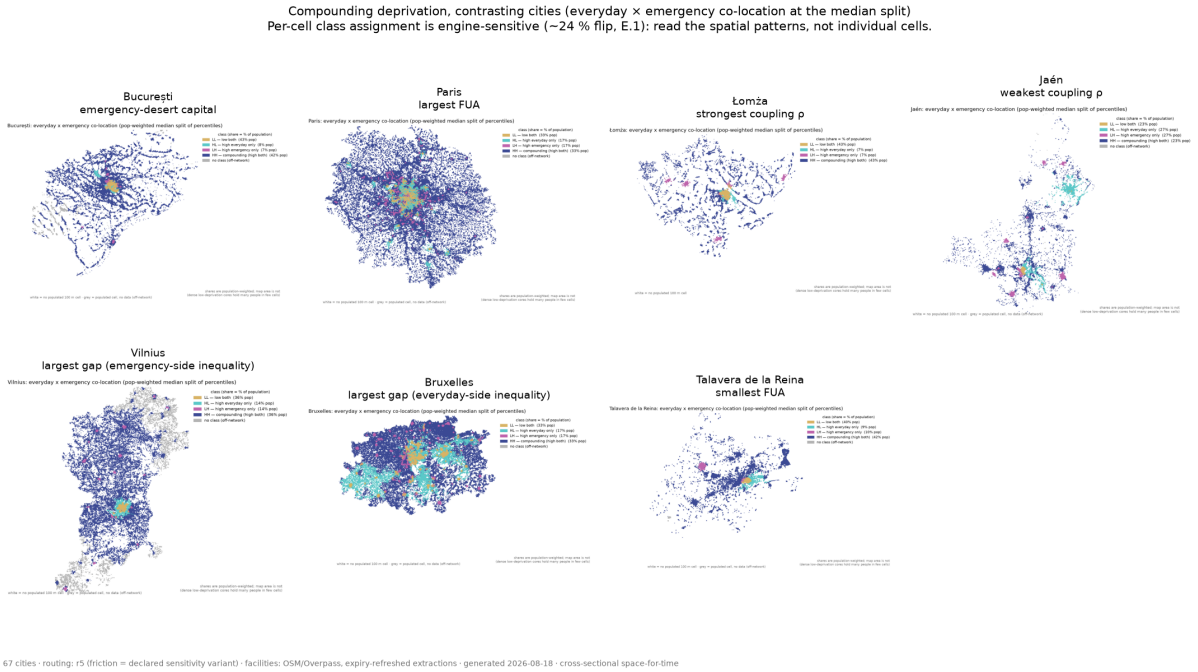


Figure 8: Compounding gallery: median-split co-location typology for contrasting cities, each panel labelled with the role it illustrates (all 67 maps in the SI). Colours: LL both low, HL high everyday only, LH high emergency only, HH compounding. Shares in the legends are population-weighted; the maps are area-weighted, so a sparse periphery dominates visually at a modest population share. Maps are read for spatial pattern only: roughly 24 % of population flips class between routing engines, so no per-cell class is a finding.

access only [41]. The theory also explains why the two regimes are different objects. A bounded level for substitutable needs and an unbounded cost for time-critical needs are what deprivation theory implies for the two kinds of good, and the clinical response literature [10, 25] supplies the anchors that transport economics' value of time, estimated on reschedulable trips, cannot. The framework (forms transferred from an estimated literature, curvature calibrated to domain anchors, robustness of ranks separated from the meaning of levels) travels to any domain that possesses anchors of comparable strength.

3.2 What urban growth delivers, regime by regime

Scaling theory's power rests on exponents that hold across cities, systems and quantities, with infrastructure in a single sublinear universality class [17, 18]; its recent refinements condition exponents on the stage of a national urban system [34]. Our results add a different kind of condition: the exponent depends on the regime of the service class, within one urban system and one harmonised measurement. Everyday deprivation scales (-0.196, explaining 44 % of cross-city variance); emergency deprivation does not (bounded within ± 0.13); and the two slopes differ within cities ($p_{\text{wild}} = 0.002$). A single infrastructure exponent is therefore not a sufficient statistic for what city size does to service deprivation, and readings of agglomeration as generalised provisioning gain overstate what growth delivers. In a mature urban system, where large cities no longer grow faster than small ones [34], what size still delivers is lower everyday deprivation. The inequality result cuts deeper: everyday inequality rises with size under every parameterisation tested, so the same agglomeration that lowers mean everyday deprivation widens its spread. Scaling theory has largely concerned means; these results argue that the scaling of within-city distributions, and specifically of welfare-weighted distributions, is where the next generation of urban scaling laws has to live. The 15-minute-city paradigm [2, 3] emerges recast: its threshold

anchors the everyday regime well, and precisely because that regime saturates, a proximity paradigm is an everyday-services paradigm, silent about the escalating-cost regime and about the compounded populations where both fail.

3.3 Built for the everyday, tested by the emergency

Resilience research has moved toward defining resilience as equitable access to essential services [26, 27], has documented that access equality and network resilience reinforce each other [42], and has shown everyday vulnerability failing to predict crisis capability [28]. Our two-regime measurement gives that distinction quantitative form across an urban system. The everyday and emergency surfaces are distinct objects with different scaling laws and different geographies: the first is set by the city and its density, the second by the coverage of a national emergency system, so a city's resilience cannot be read from either surface alone, and a city built for the everyday regime is not thereby resilient. The compounding result identifies the case the coupled-systems argument predicts: the same neighbourhoods in the periphery of both systems, above statistical independence in 66 of 67 cities, strongest where emergency inequality is highest (Central-Eastern Europe), and consistently the neighbourhoods with many children. Compounding intensity, with its fixed anchors under independence and perfect coupling, is offered as a general-purpose measure of that coupling for any pair of capability surfaces. The measurement is of the emergency regime under normal conditions, the floor from which any disruption starts. Frameworks that predict the loss of access to essential services after a hazard, building by building, take a pre-hazard baseline of network distance to the nearest operational facility as their starting point [41]; the emergency-deprivation geography measured here is that baseline for urban Europe, in welfare terms. That floor matters more each year: exposure to extreme heat in Europe is projected to rise from ten million to almost 300 million people a year [5], heat mortality to triple [4], and the Preparedness Union Strategy [11] now asks member states to plan hospitals, schools and transport for the emergency regime explicitly. A geography of emergency deprivation measured before the disruption is the baseline such planning needs.

3.4 Implications for planning and preparedness

The practical reading is short, because the two regimes point at two different jurisdictions. Everyday inequality is a big-city problem: it rises with size under every parameterisation, so the cities gaining everyday proximity on average are the ones whose internal spread widens, and intra-urban service planning is the lever. Emergency deprivation is a national coverage problem: it follows coverage grade rather than size, differs by country and macro-region, and its extreme cases are five capital cities whose peripheries no municipal instrument can serve alone. The coverage ordering gives national emergency-system planners a screening variable that access averages do not provide, and the Preparedness Union Strategy gives it an address. The consistent carriers of compounded deprivation at the European level are families with children, in 88 % of the cities studied, which makes school- and family-anchored service planning a compounding remedy rather than a sectoral one.

3.5 Limitations

Every size gradient is cross-sectional; the space-for-time reading was flagged once and applies throughout. The analysis covers walking for everyday services and driving for emergency care; public transit is deliberately out of scope, and the everyday regime is walk-only by design. Free-flow civilian drive times anchor the emergency cost scale and are not literal response-time predictions. The bias has a known direction: congestion lengthens drive times most in the largest and densest cities, so free-flow routing understates emergency deprivation there and pushes the emergency size elasticity toward a size benefit; the absence of a detectable

emergency gradient is therefore a conservative finding. Disruption itself is not modelled, so the emergency surface is the pre-disruption baseline that hazard-specific frameworks such as AccES [41] perturb. Census strata arrive on a 1 km grid broadcast to the 100 m analysis grid, so they are neighbourhood attributes; employment is unreported in the census grid for Germany and France, so no harmonised SES slope exists there by design. Per-cell typology classes carry a roughly 24 % routing-engine flip share and are read only as spatial patterns. Population shares beyond fixed time thresholds contain the finite-fill constant assigned to service-deprived cells and are footnoted wherever used. The emergency-side external benchmark (INKAR) is unavailable, and the German registry counts carry a verify-before-publication flag.

The deprivation layer itself is the first thing a reader should attack, and SI S2 reports the attack in full. In short: levels move with the calibration and are meant to (curvature alone moves a typical city's everyday Gini by 0.24 Gini points, which is why every slope claim is defended by the specification curve rather than one parameterisation), while ranks and classes do not (the same curvature moves the compounding share by 1.4 percentage points, against 29.8 for the threshold choice), and the two emergency escalation swaps are the scoped exceptions that change what the emergency Gini measures.

Two OpenStreetMap completeness specifics deserve statement rather than smoothing. Mapped GP density varies about 12-fold across countries in a way that reflects tagging conventions rather than health systems: the median FUA has 10.8 mapped GPs per 100k inhabitants, yet the country medians are 2.1 in Sweden, 2.3 in Portugal, 2.6 in Finland and 3.5 in Italy against 24.5 across the West, because primary care there sits in health centres OSM tags differently. Everyday deprivation levels are correspondingly overstated in those countries, so cross-country everyday *level* comparisons are not supported, and the Southern split of Section 2.3 (Spain and Greece above Italy and Portugal) is the visible consequence. The size gradients are unaffected: mapped GP density is uncorrelated with city size (log-log slope +0.09, country-clustered $p = 0.29$), every scaling claim is country-clustered, and adding mapped GP density as a control leaves the everyday elasticities in place (SI S4). Separately, eight cities have only one of the two emergency services mapped (Athina, Brăila, Helsinki, Lahti, Norrköping, Szeged, Talavera de la Reina, Žilina); the composite renormalises weights over the services present, their median mean emergency cost is 0.197 against 0.169 for the rest, the expected direction at a modest size, and none of them is in the desert group, so no headline turns on it. Table S11 reports the count per city.

4 Materials and Methods

4.1 Study design and sample

The study is a cross-sectional comparison of 67 functional urban areas (FUAs) in 24 European countries under one harmonised city definition, with trajectories read from the cross-sectional size gradient. The design follows the logic of Musso et al. [34], who show that what a size gradient means depends on comparing cities defined the same way; we use the Eurostat–OECD functional urban area, a densely populated core plus its commuting zone [30], because it is the one definition that exists for every European city and follows the labour-market extent of the city rather than an administrative line. There is deliberately no longitudinal component. Every statement about what happens "as cities grow" is the space-for-time reading of a cross-sectional elasticity and is labelled as such.

The sample is a stratified draw over four macro-regions (North, West, South, Central-Eastern Europe) and four population strata (100,000–250,000; 250,000–1 million; 1–5 million; above 5 million), seeded once and fixed before any deprivation surface was computed; 15, 18, 30 and 4 cities fall in the four strata, from 101,050 to 12.9 million inhabitants and 113.9 million residents in total (Table 1; every city in Table S11). Macro-region is a property of the country, so the 24 countries nest inside the four regions, and every cross-city test below is built around that nesting.

Table 3: Facilities routed to (medians across the 67 FUAs) and population-weighted median travel times to the nearest facilities of each service. Everyday services are reached on foot, emergency services by car. Source: `accessibility_by_service_pooled.csv`.

Regime	Service	Facilities per FUA	Median time (min)
Everyday	general practitioner	93	7.9
Everyday	pharmacy	228	4.9
Everyday	supermarket	325	4.7
Everyday	school	409	4.4
Everyday	green space	636	3.8
Emergency	emergency department	6	12.0
Emergency	ambulance station	6	12.0

4.2 Data

Population grid. The analysis unit is the populated 100 m cell of the GHS-POP R2023A population grid [43], clipped to each FUA. Cells with zero population enter no statistic; every aggregate in the paper is population-weighted over the populated cells.

Facilities. Facilities come from OpenStreetMap through Overpass queries with one harmonised tag taxonomy for all cities and an expiry-refreshed extraction date. Five everyday service categories are routed on foot (general practitioners, pharmacies, supermarkets, schools and green spaces, the last two split into a local and a district sub-type that share one extraction and weigh 0.5 each so that each category counts once) and two emergency services by car (hospitals with an emergency department, and ambulance stations). Table 3 gives the median facility counts per FUA. Completeness was benchmarked against national registries where available (Germany: pharmacies 0.976, hospitals 1.24 of the registry count) and characterised per country through the per-city facility counts, which surface the general-practitioner tagging artefact discussed in the limitations.

Street networks. Travel times are computed over the OpenStreetMap street network (Geofabrik extracts, clipped to the FUA window) with the R5 routing engine via r5py [44]. A friction-surface engine [14] was the planned primary and is retained as a declared sensitivity variant after a two-city cross-check showed city-specific level bias (SI S7).

Census strata. Vulnerability strata come from the Eurostat Census 2021 1 km grid (broad age groups, harmonised under Regulation 2018/1799 for every city) and, where a country publishes one, from national fine grids (for Germany the Zensus 2022 100 m grid with age, rent, tenure and vacancy). The 1 km census values are broadcast to the 100 m analysis cells inside each kilometre and are therefore neighbourhood attributes. Withheld census counts are treated as missing, never as zero.

4.3 Travel times

For every populated cell we compute the travel time to the 30 nearest facilities of each service, walking at 4.8 km/h for the everyday regime and driving at free-flow speed for the emergency regime, with a 30-minute cut-off on foot and a 60-minute cut-off by car. Matrices are routed from facilities to cells and transposed. For walking the network is symmetric; for driving the transposed direction is the substantively correct one for an ambulance, which travels from its station to the patient, and the measured asymmetry of the car network is a median of one minute. Free-flow civilian drive times anchor the emergency cost scale but are not response-time predictions: dispatch and turnout are unmodelled, lights-and-siren travel is faster than civilian

flow, and free-flow understates congested cores relative to the periphery, which is conservative for the periphery findings and pushes the emergency size elasticity toward a size benefit (Section 3.5). Cells that reach no facility of any service in any mode have no network path; they are masked from every surface under one shared mask, and their population share is reported. Cells that are routable but have no facility of a given service within its cut-off are service-deprived rather than unreachable: they receive a large finite time so that the everyday level saturates and the cell stays on the map. The two situations are kept apart because merging them lets one sparse service mask a whole city.

4.4 Deprivation functions

Deprivation is a function of the time a resident is without a service, and it takes a different form for the two kinds of need [19, 20]. In both regimes the functional form is transferred from the estimated literature and the curvature is calibrated to domain anchors, because the published coefficients were estimated on an hours-scale deprivation time (hours without water, food or care) that does not transfer to intra-urban minutes. The curvature parameters are the primary targets of the sensitivity analysis.

Everyday deprivation level. For each everyday service s the travel times to the reachable facilities j are first adjusted for congestion by a two-step floating catchment area factor [45, 46]: the supply-to-demand ratio of a facility, $R_j = S_j / \sum_i P_i K(t_{ij})$ with a Gaussian kernel K, is converted into a time inflation $c_j = (R_{\text{ref}}/R_j)^{\gamma}$ relative to the demand-weighted median ratio of the city, so that crowded facilities count as further away. The effective time of a cell is then a soft minimum over its reachable facilities,

$$t_{is}^{\text{eff}} = -\frac{1}{\kappa} \ln \sum_j \exp(-\kappa\, t_{ij}\, c_j), \tag{1}$$

which lies between the nearest adjusted time and that time less $\ln(n)/\kappa$: several similar options count as slightly closer than the nearest alone, which is what substitutability means. The deprivation level of the cell for that service is the logistic

$$D_{is} = \frac{1}{1 + \exp(-k\,(t_{is}^{\text{eff}} - t_0))}, \qquad t_0 = 15 \text{ min},\ k = 0.2 \text{ min}^{-1}, \tag{2}$$

the saturating S-curve of needs-based severity [21], with its inflection at the 15-minute-city threshold [2] and a ceiling of 1 that the surface reaches by about 45 minutes. The everyday deprivation level of a cell is the weighted mean of D_{is} over the service categories reachable from it. Because network walking time scales inversely with speed, the 3.5 km/h often assumed for elderly and mobility-impaired residents is equivalent to lowering t_0 from 15 to about 11 minutes at the baseline speed, which the 10-minute inflection variant of the sensitivity grid covers.

Emergency deprivation cost. The emergency surface uses the nearest facility only and a convex Box-Cox deprivation cost function [22, 24],

$$C_i = \frac{g(\min_j t_{ij})}{g(15)}, \qquad g(t) = \frac{(t+1)^{\lambda} - 1}{\lambda}, \qquad \lambda = 1.8. \tag{3}$$

The anchors are the emergency-medical-service response benchmarks: the cost is negligible below the 3–4 minute ideal [31, 47], about a third of the benchmark cost at the 8-minute urban standard, which the clinical literature contests (no survival effect for most patients in [31], supported in [32]; attributed via [48]), and equal to the benchmark cost at the 15-minute upper target [33, 48]. Convexity is supported independently by the distance-mortality evidence

[25]. Solving λ from the anchors gives 1.89–1.95, inside the sensitivity sweep of 1.4–2.2, so the transferred and the anchor-calibrated curvature coincide. The 8-minute mark cannot play the half-maximum role the 15-minute inflection plays for the everyday level: convexity caps $g(8)/g(15)$ at 0.427 given $g(4)/g(15) = 0.10$, so the anchor structures of the two regimes are genuinely different. Dividing by $g(15)$ reports every cost in multiples of the deprivation of a resident whose ambulance arrives at the benchmark; it does not bound the function, and because it is division by a constant it leaves every rank-based statistic and both Ginis exactly unchanged.

4.5 City-level indicators and cross-regime comparison

The everyday level and the emergency cost are different mathematical objects on incomparable scales, and the analysis is built so that only unit-invariant statistics cross between them. Three families qualify. Elasticities (slopes of a log outcome on log population) are invariant to any rescaling of a surface. The within-city Gini,

$$G = \frac{\sum_i \sum_j P_i P_j \, |x_i - x_j|}{2 \, (\sum_i P_i)^2 \, \bar{x}}, \tag{4}$$

computed on the raw values of one surface with population weights P_i, is mean-normalised and therefore unit-free, so comparing the inequality of a level with the inequality of a cost is the same move as comparing an income Gini with a wealth Gini. Rank-based statistics are invariant to any monotone transform. The rank layer is the population-weighted mid-rank percentile of each surface, which places a group of tied cells at the midpoint of its own weight span and tames the unbounded emergency tail. Raw magnitudes are never summed, differenced, co-plotted on one axis or clustered across regimes.

The co-location typology cuts both percentile surfaces at their population-weighted medians. Because each marginal is then one half by construction, the 2×2 table of low-low, high-low, low-high and high-high cells has a single free number, the high-high (HH) compounding share: HL = LH = 0.5 − HH and LL = HH. Statistical independence of the two surfaces gives HH = 0.25, so the excess over one quarter is the co-location beyond chance. Alongside the thresholded share we report the Spearman coupling ρ of the two percentile surfaces and the threshold-free compounding intensity, the population-weighted mean of $\min(p_i^{\mathrm{ev}}, p_i^{\mathrm{em}})$ over cells, which equals 1/3 under independence, 1/2 under perfect coupling and 1/4 under perfect divergence. The divergence gap, the emergency Gini minus the everyday Gini, is the one cross-regime subtraction; it is a difference of dimensionless indices, depends on the functional forms, and never carries a headline alone.

4.6 Vulnerability strata

Vulnerability is analysed on two strictly separated levels. The census-harmonised age strata (top quartile of the share of residents under 15, top quartile of the share aged 65 and over) exist for every city and are pooled across cities. National fine-grid strata (lowest rent quartile, lowest ownership quartile, finer age cuts) exist only where a national grid does, use different definitions, and are reported per city and never pooled. Each stratum's mean deprivation and compounding share are compared against the cells on which its own census variable is published, because national grids are published where the population is large enough and that is not a random subset of a metropolitan area; a stratum whose variable covers less than half of a city's population is dropped from the pooled summary. Within-stratum inequality uses the concentration index [49].

4.7 Cross-city analysis and inference

Size regressions. Following the scaling tradition [17], each city outcome y_c is regressed on population,

$$\ln y_c = \alpha + \beta \ln P_c + \varepsilon_c, \tag{5}$$

so that β is an elasticity. Because cities nest in 24 countries and every outcome has a national component, the citable inference clusters standard errors on country and reports a wild cluster bootstrap p-value with the null imposed, Rademacher weights per country and 4,999 draws [50, 51]; heteroskedasticity-robust and country-fixed-effect estimates are reported as references. The claim that emergency deprivation does not scale is stated positively through two one-sided equivalence tests [37] at bounds of ± 0.10 and ± 0.13. The difference between the two regimes' slopes is tested within cities by stacking both outcomes and estimating a regime $\times$ log-population interaction with the same clustered inference. Every headline elasticity carries a leave-one-out envelope, the Cook's-distance top city, Huber and median-regression re-estimates, and re-estimation without the five desert capitals.

Regional contrasts. Region is a property of countries, so a regional contrast is tested by comparing country means and permuting whole countries across regions with the regional country counts held fixed (10,000 permutations); the city-level analysis of variance is printed only as an anti-conservative reference, and a country-random-intercept mixed model triangulates. Three further analyses characterise where the cross-city variance lies (SI S3). A nested sum-of-squares decomposition splits each indicator's variance into between-region, between-country-within-region and within-country shares. Dispersion by region is compared with a Fligner–Killeen test whose statistic is re-evaluated under the same country permutation. Every regional contrast is re-run without the five desert capitals.

Clustering and coverage grades. The city vector (means, Ginis, coupling, compounding, tail shares, within-city density slopes, log population) is robust-scaled, with the zero-inflated tail shares opened by $\log(1 + x)$ so the scaler cannot manufacture separation, and clustered by k-means and by Ward's method with k selected by silhouette over 2–6. Every partition carries the silhouette [40], a bootstrap adjusted Rand index over resampled cities [39], a Gaussian silhouette null built from same-covariance multivariate-normal draws pushed through the same selection, and the adjusted Rand index against the region partition, since with country nesting a cluster can be a region in costume. When the winning split isolates an outlier group below a quarter of the sample, a peeled second pass re-scales and re-clusters the remaining cities with the same diagnostics. The coverage grade of a city is read off the two passes and never from a hand-set threshold: desert is the main pass's outlier group, partial desert the peeled pass's smaller cluster, covered the rest. A feature-set check re-runs the whole procedure with the 8- and 15-minute benchmark shares added. The graded scaling model regresses log emergency cost on centred log population, grade dummies and their interaction with country-clustered errors; centring makes the dummy test a contrast between grades at a typical city size.

Deprivation against access. Every headline size regression is re-run on the deprivation-free travel-time indicators the accessibility literature would use (median, mean and, for the emergency regime, the 90th-percentile minutes), the deprivation-based and time-based Gini families are correlated across cities, and the desert capitals' ratios to the sample are computed under both measures. The everyday 90th percentile is not regressed, because above the walking cut-off the everyday composite time behaves as a count of unmet service categories rather than a travel time.

Additional robustness. Because the everyday and emergency regressions share the same regressor, joint (seemingly unrelated) estimation coincides with equation-by-equation least squares; what it would add is the cross-equation residual correlation, which we report. Residual spatial dependence is tested by Moran's I over the city-centre coordinates (five nearest neighbours, row-standardised weights, permutation p). The elasticities are re-estimated with mapped general-practitioner density as a completeness control, with a centred quadratic in log population with and without coverage-grade dummies, and with size-stratum dummies in place of log population, and the minimum detectable emergency elasticity at 80 % power is derived from the clustered standard error (SI S4).

4.8 Robustness of the deprivation layer

A structured, non-probabilistic sweep recomputes the rank-based targets over a defensible envelope and reports how far they move (SI S2). Layer 1 varies the curvature within the fixed forms ($k \in \{0.1, 0.2, 0.3\}$ per minute and $t_0 \in \{10, 15, 20\}$ minutes for the everyday logistic; $\lambda \in \{1.4, 1.8, 2.2\}$ for the emergency Box-Cox). Layer 2 swaps the forms while pinning the same anchors: a concave Box-Cox everyday level through the same half-maximum and ceiling, and for the emergency regime an exponential escalation [19] and a bounded survival-based curve, both solved from the 8- and 15-minute anchors, so that the three emergency forms coincide inside the benchmark window and diverge only beyond it, where no benchmark exists. Layer 3 sweeps the accessibility model (soft-minimum κ, congestion exponent γ, catchment bandwidth, number of nearest facilities, everyday mode set, unreachable fill, off-network policy) from the saved travel-time matrices. A threshold axis sweeps the typology cut from the 40th to the 75th percentile. Results are reported as rank agreement of city orderings against the baseline, min–max envelopes per city and axis, class-share envelopes, the share of population whose typology class flips, and the specification curve of the inequality slopes over all 14 deprivation parameterisations [38].

4.9 Reproducibility and data availability

The pipeline is configuration-driven with provenance sidecars on every derived travel-time matrix, and no deprivation parameter is hard-coded: every value lives in the configuration with its literature source. All tables and figures trace to the persisted 67-city state, and every number quoted in the text is checked against its source table by an automated audit of 190 checks run in continuous integration; the main-text figures are drawn from the pack tables and the configuration by scripts in the study repository. Code, configuration, the full result tables, the per-city maps and the sensitivity sweeps are available in the study repository and its results branch. Population grid: GHS-POP R2023A [43]. Facilities and street networks: OpenStreetMap. Census strata: Eurostat Census 2021 grid and the German Zensus 2022 grid. City boundaries: Eurostat–OECD functional urban areas.

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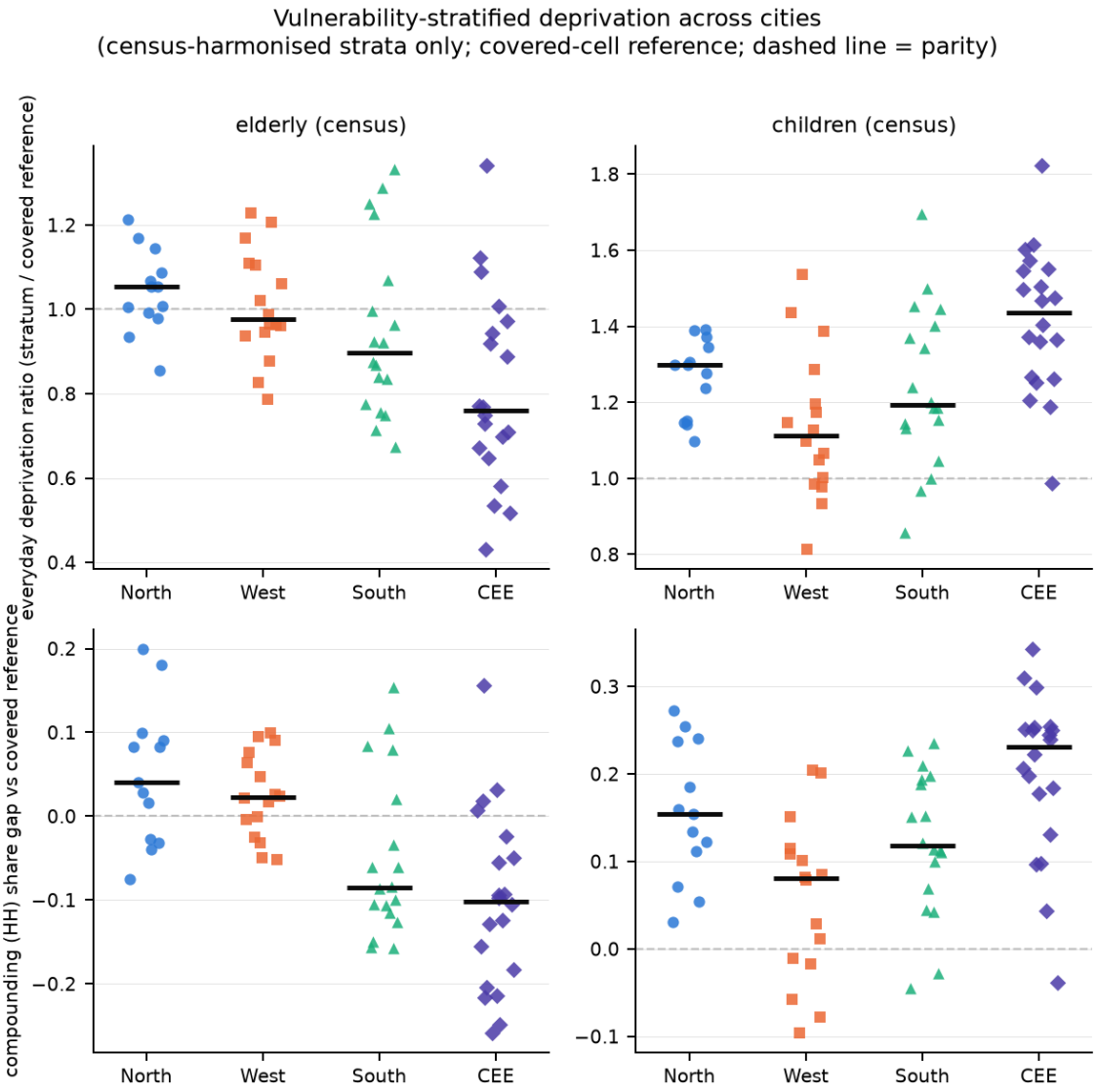


Figure 9: Who carries the deprivation: census-harmonised vulnerability strata by macro-region, pooled across the 67 cities against covered-cell references (dashed line = parity; bar = regional median). Columns: elderly and children strata; top row: everyday deprivation ratio (stratum vs reference); bottom row: compounding (HH) share gap. Children (top-quartile child-share cells) experience higher everyday deprivation in 88 % of cities (median ratio 1.27) and higher compounding in 88 % (median HH gap +0.13), strongest in CEE (1.44) and the North. The elderly stratum sits at parity (median 0.96, above 1 in 37 % of cities); no European elderly claim is made. National fine-grid strata are per-city depth and are never pooled with these.

Supplementary Information

The two geographies of urban deprivation

S1 The robustness architecture in one view

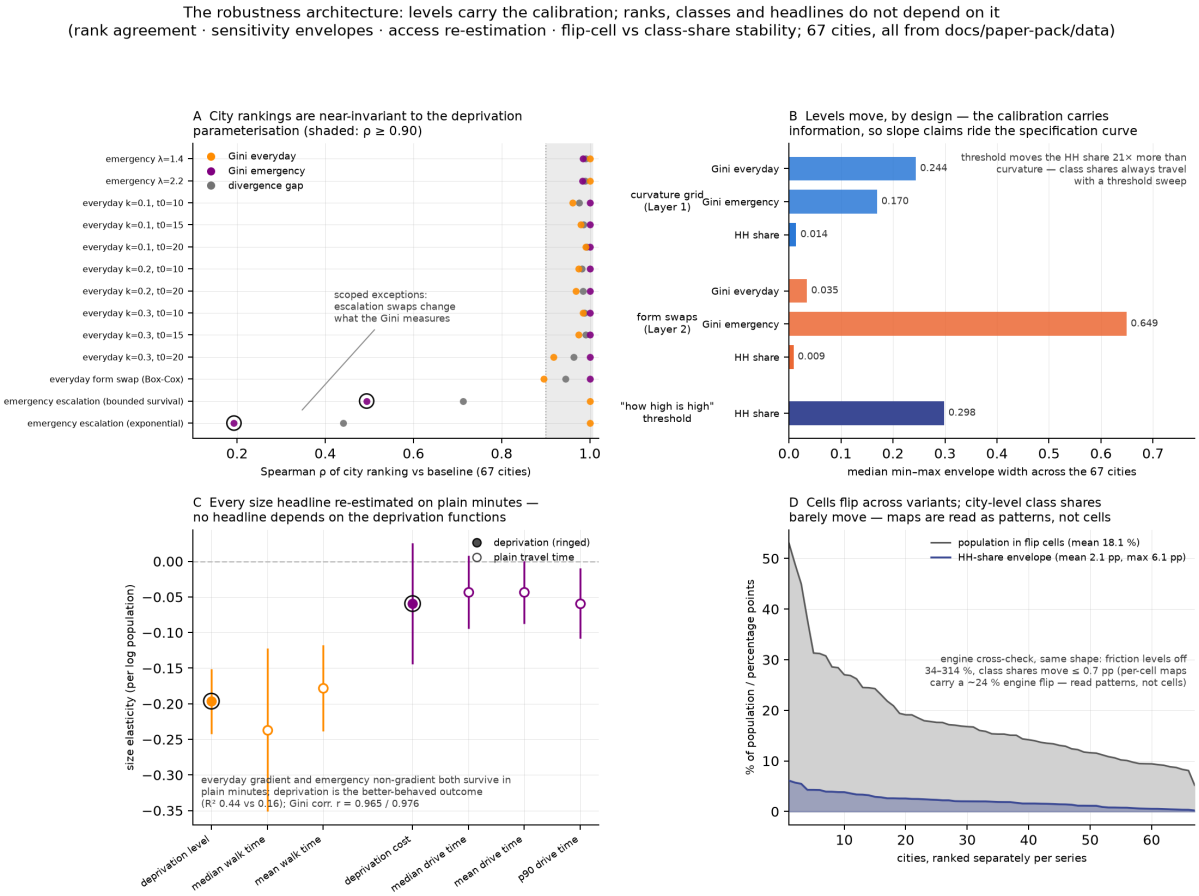


Figure S1: The robustness architecture (contribution claim 5). One view of the checks defending the results: rank agreement of city orderings across deprivation variants (min $\rho = 0.90$ over the curvature grid and everyday form swap; the two emergency escalation swaps are scoped exceptions that change the measurand, $\rho = 0.19$ exponential, 0.49 bounded survival); sensitivity envelope widths per axis (Table S1); access-based re-estimation of every headline (Table S9); and flip-cell against class-share stability (population-weighted flip share 18.1 % on average, 5.2–53.0 % across cities, against HH class-share movement of 2.1 pp average, 6.1 pp maximum), with the routing-engine cross-check quoted alongside (engine flip ≈ 24 %, aggregate class shares ≤ 0.7 pp).

Rank-based results are near-invariant to the deprivation parameterisation; level and inequality results move with the calibration, which is where the calibration carries information. The flagged exceptions are the two emergency escalation swaps, each of which changes what the emergency Gini measures (an extreme-tail-depth ordering under exponential escalation; a benchmark-window inequality ordering under the bounded survival curve), and they are reported as changes of measurand.

S2 Robustness of the deprivation layer

A paper whose contribution is a measurement layer must expect that layer to be attacked first, and the answer has two halves that both need stating. What is varied is drawn in Fig. 1: the

Table S1: Sensitivity envelopes of the deprivation layer: median min–max envelope width across the 67 cities, per sweep axis and target. Curvature moves the Ginis (the calibration carries information); no axis of the calibration moves the compounding share materially compared with the "how high is high" threshold, which dominates by a factor of about 21. The form-swap emergency-Gini envelope (0.649) spans the three beyond-cut-off escalation hypotheses and is a boundary-of-validity statement, reported with H2's conditionality in Section 2.2. Source: `deprivation_sensitivity_summary.csv`.

Sweep axis	Everyday Gini	Emergency Gini	HH share
Curvature (Layer 1)	0.244	0.170	0.014
Form swap (Layer 2)	0.035	0.649	0.009
"How high is high" threshold	–	–	0.298

Layer-1 curvature grid ($k \in \{0.1, 0.2, 0.3\}$ per minute $\times$ $t_0 \in \{10, 15, 20\}$ minutes for the everyday logistic; $\lambda \in \{1.4, 1.8, 2.2\}$ for the emergency Box-Cox) and the Layer-2 form swaps calibrated to the same domain anchors (a concave Box-Cox everyday level; exponential and bounded survival-based emergency escalations), all against the linear loss of a pure-access average.

How far the results move is in Table S1. Levels are not robust, and are not meant to be: curvature alone moves a typical city's everyday Gini by 0.24 Gini points (median envelope width across the 67 cities), which is why the calibration carries information and why every slope claim is defended by the specification curve rather than by one parameterisation. Ranks and classes are robust: the same curvature grid moves the compounding share by 1.4 percentage points at the median, the form swaps by 0.9, against 29.8 for the "how high is high" threshold. City-level Gini orderings survive with two scoped exceptions: rank agreement with the baseline stays at $\rho \geq 0.90$ over the curvature grid and the everyday form swap, while both emergency escalation swaps reorder the emergency-Gini ranking (exponential $\rho = 0.19$; bounded survival $\rho = 0.49$) precisely because each changes what that Gini measures, a change of measurand reported as such rather than noise.

The wider inventory reads as one block. Deprivation-variant flip cells, the population whose typology class changes under any deprivation parameterisation, average 18.1 % of city population (range 5.2–53.0 %), with the HH class share moving 2.1 percentage points on average and 6.1 at most. The clustering survives its own feature-set choice (main partition ARI 1.00, peeled 0.97 with benchmark shares included). The routing-engine cross-check bounds the engine's leverage: switching engines moves travel-time levels by 34–314 % and flips the class of roughly 24 % of population per city, while aggregate class shares move at most 0.7 percentage points; per-cell map readings are therefore never findings, aggregate shares are. OSM completeness was benchmarked against national registries where available (Germany: pharmacies 0.976, hospitals 1.24) and the external INKAR benchmark passed. Fig. SS1 assembles the architecture in one view. The reading discipline it supports ran through the whole paper: rank-based results are near-invariant to the deprivation parameterisation, and level and inequality results are where the calibration carries the information.

S2.1 What the results change for theory

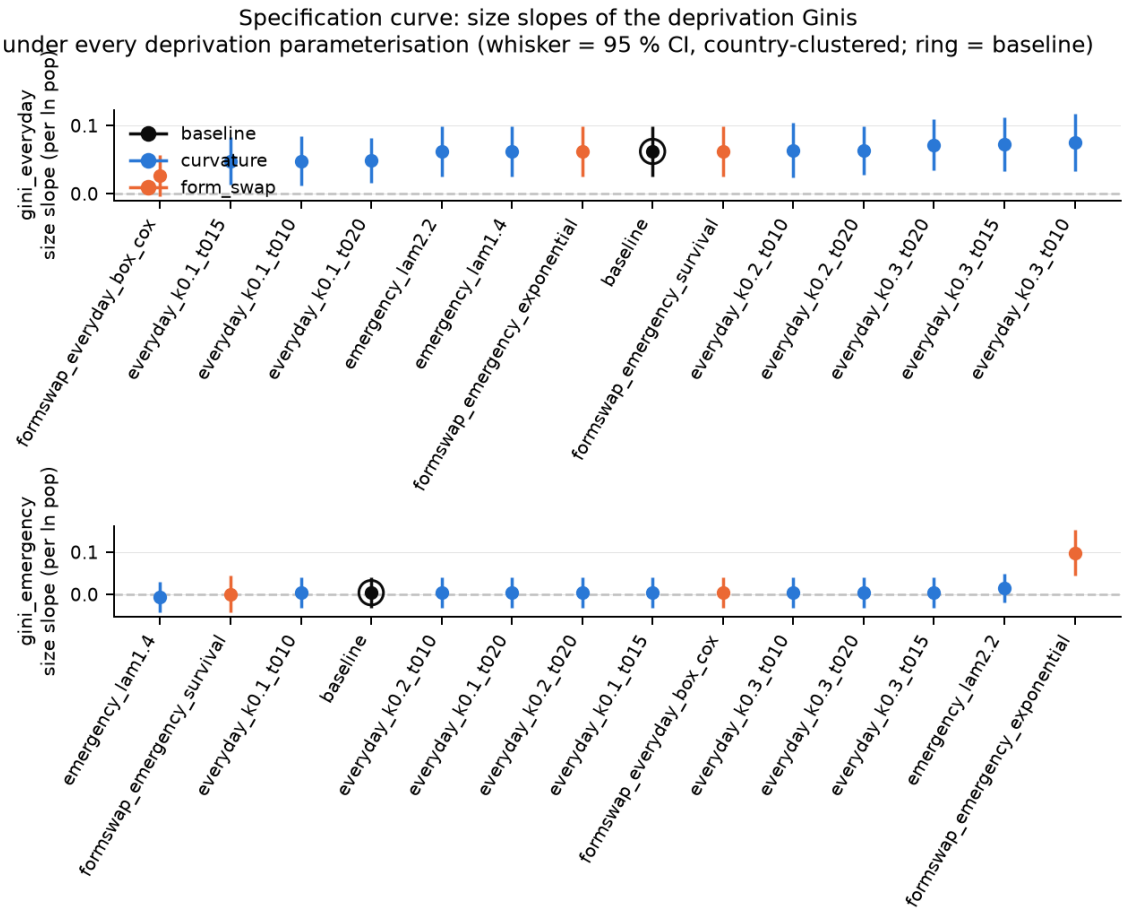


Figure S2: Specification curve of the inequality-scaling slopes. Top: everyday Gini; bottom: emergency Gini. Each point is the size elasticity of the within-city Gini re-estimated under the ringed baseline or one of the 14 alternative deprivation parameterisations (blue: curvature grid; orange: functional-form swaps); whiskers are country-clustered 95 % CIs. The everyday-Gini slope is positive under every specification (+0.026 to +0.075; significant in 13/14, the exception being the concave Box-Cox form swap at $p = 0.083$). The emergency-Gini slope splits by beyond-cut-off escalation form: flat under saturating and polynomial escalation, rising only under exponential escalation (+0.098, $p = 0.0003$).

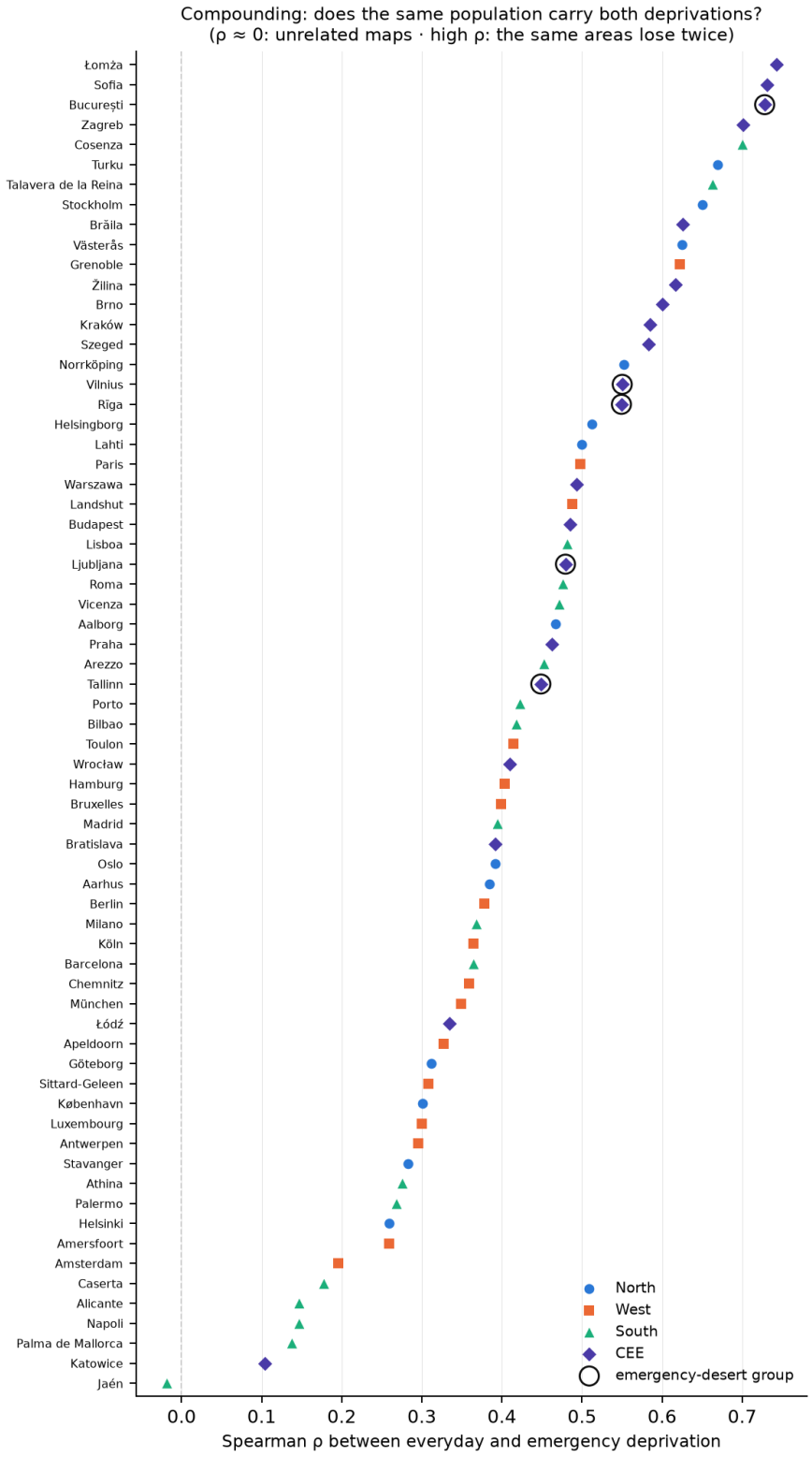


Figure S3: Coupling of the two deprivation surfaces across the 67 cities. Spearman ρ between the everyday and emergency population-weighted percentile surfaces, cities ranked; ρ spans -0.02 to 0.74. Colours mark macro-regions; the emergency-desert capitals are outlined. The HH compounding share exceeds the 25 % independence benchmark in 66 of 67 cities (sample mean 33 %). Rank-based measures are invariant to the deprivation parameterisation.

S3 Regional structure: variance, dispersion, deserts excluded

Table S2: Nested decomposition of the variance of each city indicator across the 67 cities: share of the total sum of squares lying between the four macro-regions, between countries within regions, and between cities within countries. Source: `regional_variance_decomposition.csv` (`paper/analysis_regional_structure.py`).

Indicator	Between regions	Between countries (within region)	Within countries	Country level (regions + countries)
coupling ρ	0.20	0.22	0.58	0.42
everyday Gini	0.24	0.51	0.25	0.75
emergency Gini	0.35	0.35	0.30	0.70
divergence gap	0.32	0.44	0.24	0.76
HH share	0.22	0.18	0.60	0.40
compounding intensity	0.21	0.19	0.60	0.40
mean emergency cost	0.25	0.54	0.22	0.78

Table S3: Dispersion of each indicator by macro-region (standard deviation across cities) with a Fligner–Killeen test of equal dispersion at the city level (anti-conservative reference) and the country-permutation p-value of the same statistic (whole countries permuted across regions, 10,000 permutations). Source: `regional_dispersion.csv`.

Indicator	sd North	sd West	sd South	sd CEE	p city	p_{perm}
coupling ρ	0.144	0.102	0.187	0.154	0.143	0.163
everyday Gini	0.075	0.079	0.102	0.060	0.067	0.165
emergency Gini	0.058	0.065	0.076	0.130	0.006	0.028
divergence gap	0.074	0.072	0.102	0.159	0.008	0.010
HH share	0.038	0.026	0.044	0.040	0.278	0.241
compounding intensity	0.017	0.013	0.023	0.020	0.175	0.190
mean emergency cost	0.481	0.297	0.679	1.294	0.035	0.063

Table S4: Regional contrasts re-tested without the five emergency-desert capitals (62 cities, 24 countries): ANOVA on country means and the country-permutation p-value, with the Central-Eastern European and rest-of-sample medians. Source: `inference_regional_no_deserts.csv`.

Indicator	p country ANOVA	p_{perm}	Median CEE	Median rest
coupling ρ	0.0028	0.0030	0.583	0.384
everyday Gini	0.2486	0.2613	0.522	0.458
emergency Gini	0.0080	0.0060	0.507	0.437
divergence gap	0.0364	0.0346	-0.004	-0.054
HH share	0.0012	0.0016	0.359	0.319
compounding intensity	0.0021	0.0023	0.397	0.374
mean emergency cost	0.0170	0.0198	1.441	1.062

Table S5: Coverage grade by macro-region (number of cities). Source: `coverage_grade_by_region.csv`.

Region	Covered	Partial desert	Desert
North	6	7	0
West	15	1	0
South	17	1	0
CEE	10	5	5

Table S6: The 18 Southern cities ordered by everyday Gini: the Spanish and Greek cities lie above the Italian and Portuguese ones with no overlap. Source: `south_everyday_gini_by_country.csv`.

City	CC	Population	G_{ev}	$\bar{D}_{\text{ev}}$
Athina	EL	3,522,264	0.609	0.103
Bilbao	ES	1,053,444	0.570	0.160
Barcelona	ES	5,945,709	0.545	0.175
Jaén	ES	249,477	0.543	0.291
Madrid	ES	7,601,394	0.533	0.191
Alicante	ES	996,867	0.530	0.311
Talavera de la Reina	ES	101,050	0.522	0.378
Palma de Mallorca	ES	818,202	0.503	0.251
Roma	IT	4,432,845	0.424	0.287
Arezzo	IT	146,788	0.412	0.381
Cosenza	IT	239,305	0.407	0.505
Milano	IT	5,030,024	0.400	0.183
Vicenza	IT	280,054	0.391	0.275
Palermo	IT	1,006,055	0.383	0.317
Lisboa	PT	3,132,904	0.367	0.239
Napoli	IT	3,457,649	0.305	0.326
Caserta	IT	298,402	0.298	0.283
Porto	PT	1,531,690	0.272	0.324

S4 Additional robustness of the size regressions

Table S7 re-estimates the four headline size regressions with checks suggested by an independent methods review. Residual spatial dependence is absent for the everyday level (Moran's $I = 0.06$, permutation $p = 0.41$) and present for the emergency level ($I = 0.23$, $p = 0.0015$) and for both Ginis ($I = 0.20$ and 0.39): cities close to one another share what size does not explain, which is the country and coverage structure of Section 2.3 (Table S2). The slope estimates are unaffected by that dependence; the inference on them is country-clustered throughout, and regional contrasts are tested by permuting countries, so the dependence enters the tests at the level at which it arises. Adding mapped GP density per 100k inhabitants, the completeness proxy of the limitations, as a control leaves the everyday elasticity at -0.185 (wild $p = 0.0002$) and the everyday-Gini elasticity at $+0.051$ ($p = 0.0006$); the control itself is significant in both everyday equations (more mapped practices, lower mean deprivation and higher within-city inequality) and in neither emergency equation. A centred quadratic in log population is not significant for the everyday level or for either Gini, and is significant for the emergency level (-0.169, wild $p = 0.001$; -0.104, $p = 0.010$ with coverage-grade dummies): the size-stratum means of the emergency deprivation cost are 1.15, 1.69, 1.56 and 0.70 multiples of g(15) from the smallest to the largest stratum, an inverted U in which the middle strata, where the desert and partial-desert cities sit, carry the highest cost and the four FUAs above five million the lowest. With four cities in the top stratum we report the shape and do not headline it; the linear statement of Section 2.2 stands, with a minimum detectable emergency elasticity of 0.12 at 80 % power, in line with the equivalence bound of ± 0.13. Table S8 gives the cross-equation residual correlation: with identical regressors in both regime equations, joint (seemingly unrelated) estimation coincides with equation-by-equation OLS, and what it would add is the correlation of the residuals, $r = 0.51$ for the means (cities with more everyday deprivation than their size predicts also have more emergency deprivation, the city-level face of compounding) and $r = 0.19$ for the Ginis ($p = 0.13$). The two Ginis themselves are uncorrelated across cities ($r = 0.12$, $p = 0.32$).

Table S7: Additional robustness of the four log-log size regressions (67 cities, country-clustered). Moran's I of the residuals over the city-centre coordinates (5 nearest neighbours, row-standardised; expected value -0.015; permutation p, 9,999 draws); the elasticity with log mapped GP density per 100k inhabitants added as a control (wild cluster bootstrap p for the elasticity; clustered p for the control); the coefficient on the squared centred log population (wild p); the country-clustered Wald p of three size-stratum dummies replacing log population; and the minimum detectable elasticity at 80 % power (2.80× clustered SE). Source: `scaling_robustness_extra.csv` (`paper/analysis_scaling_robustness.py`).

Outcome	Elast.	Moran I	p_{perm}	Elast. GP	p_{wild}	GP coef. (p)	Quadr. (p_{wild})	Strata p
mean everyday level	-0.196	+0.061	0.406	-0.184	0.0002	-0.125 (0.00)	-0.023 (0.24)	0.000
mean emergency cost	-0.059	+0.232	0.001	-0.048	0.2838	-0.115 (0.18)	-0.169 (0.00)	0.000
everyday Gini	+0.062	+0.197	0.005	+0.051	0.0004	+0.121 (0.00)	+0.003 (0.76)	0.052
emergency Gini	+0.004	+0.388	0.000	+0.006	0.7998	-0.018 (0.47)	+0.004 (0.82)	0.759

Table S8: Cross-equation and cross-regime correlations. Residual correlation between the everyday and the emergency size regression (with identical regressors, joint SUR estimation coincides with OLS and this correlation is what joint estimation adds) and the cross-city correlation of the two within-city Ginis. Source: `scaling_crossequation.csv`.

Pair	r	p
residuals of the mean equations	+0.511	0.0000
residuals of the Gini equations	+0.186	0.1322
everyday Gini vs emergency Gini (Pearson)	+0.123	0.3202
everyday Gini vs emergency Gini (Spearman)	+0.117	0.3439

S5 Size gradient of the divergence measures

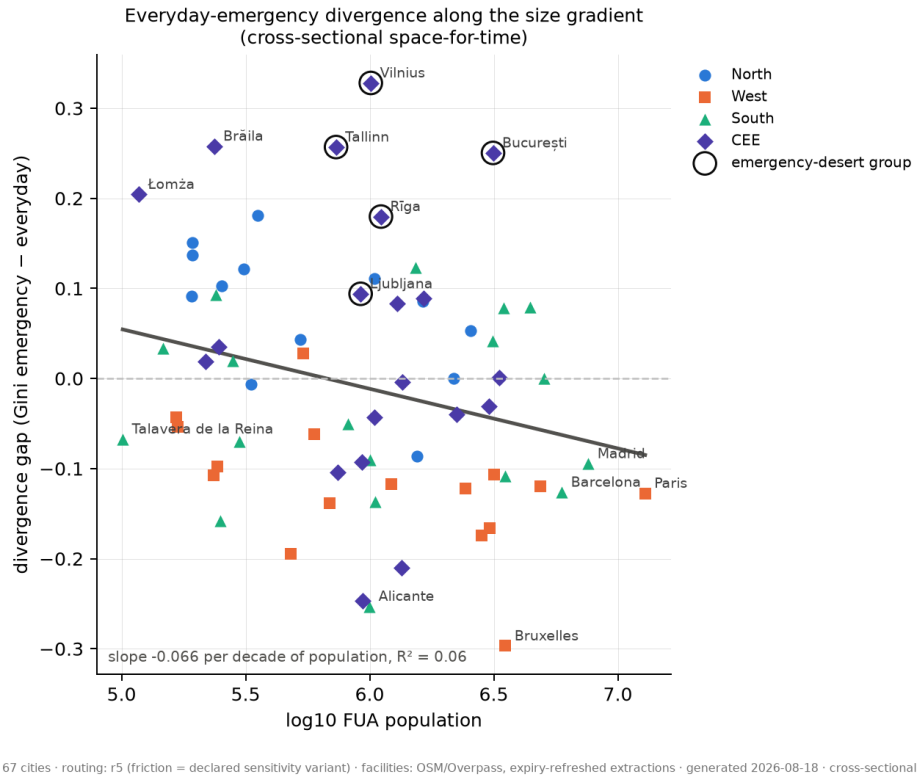


Figure S4: The divergence gap (emergency Gini minus everyday Gini) against $\log_{10}$ FUA population, coloured by macro-region with the emergency-desert capitals outlined. The gap falls weakly with size (-0.066 per decade of population, $p = 0.012$, $R^2 = 0.06$). Size is a poor predictor of where the two regimes diverge; the structure is regional and coverage-graded (main text Section 2.3).

The companion gradients are weaker still: the coupling ρ ($p = 0.14$) and the compounding share ($p = 0.06$) show no significant size gradient (`size_gradient.csv`).

S6 Deprivation vs access: exact values

Table S9: Size elasticities of deprivation-based and minutes-based outcomes (log-log, country-clustered SEs, 67 cities; cross-sectional space-for-time), plotted in main-text Fig. 2A. The everyday p90 is not regressed: above the walk cutoff the everyday composite time behaves as a count of unmet service categories rather than a travel time. Source: `deprivation_vs_access.csv`.

Regime	Outcome	Elasticity	p	R^2
Everyday	mean deprivation level	−0.196	< 0.001	0.44
Everyday	median walk time	−0.237	< 0.001	0.16
Everyday	mean walk time	−0.178	< 0.001	0.31
Emergency	mean deprivation cost	−0.059	0.16	0.02
Emergency	median drive time	−0.043	0.09	0.04
Emergency	mean drive time	−0.043	0.05	0.04
Emergency	p90 drive time	−0.059	0.02	0.05

S7 Validation exercises

E.1 Routing-engine cross-check. Re-running full cities under the friction-surface engine against the R5 baseline (Hamburg, Köln) shows travel-time levels understated by 34–314 % with city-specific Gini bias (emergency Gini −30 % in Hamburg, −18.5 % in Köln), which demoted friction to sensitivity variant. Aggregate typology class shares move ≤ 0.7 pp and the divergence ρ by ≤ 11 %, while roughly 24 % of population flips per-cell class; hence per-cell map readings are never findings.

E.2 OSM completeness. Country-level benchmark against national registries where available: Germany, pharmacies ratio 0.976, hospitals 1.24 (the hospital ratio carries a verify-before-publication flag on the registry side). The per-city facility counts (`accessibility_by_service_cities.csv`) provide the cross-country completeness picture the single-country registry benchmark cannot, and surface the GP tagging artefact discussed in the limitations.

E.3 External benchmark. The German INKAR accessibility indicators reproduce the study's German pharmacy and supermarket access patterns (passed); no comparable emergency-side benchmark is available.

E.4 Resolution and E.5 face validation. Quantile-curve comparisons of regime times under both engines and per-anchor-city face validation documents are maintained in the study repository.

Re-anchor verification. The emergency reporting anchor moved from $g(45)$ to $g(15)$ in August 2026. For the three cities re-seeded from stored artifacts the verification is exact: every rank-based indicator is identical to machine precision (max $|\Delta| \approx 10^{-16}$) and the level ratio equals $g(45)/g(15) = 6.7307$ to 10^{-15}; the other 64 cities were byte-identical between the two runs.

S8 Indicator glossary

Table S10: Compact glossary of the indicators used in this paper.

Term	Meaning
FUA	Functional urban area (Eurostat/OECD): city plus commuting zone; the unit "city".
Everyday regime	Five walkable service categories (GP, pharmacy, supermarket, school, green space); deprivation is a bounded *level* in $[0, 1]$.
Emergency regime	ED hospital and ambulance station, reached by car; deprivation is an unbounded convex *cost* in multiples of g(15).
$\bar{D}_{\text{ev}}$, $\bar{C}_{\text{em}}$	Population-weighted mean deprivation per regime; scales are incomparable across regimes.
G_{ev}, G_{em}	Population-weighted within-city Gini of each surface.
Divergence gap	$G_{\text{em}} - G_{\text{ev}}$; positive = emergency-side inequality problem.
ρ	Spearman correlation of the two population-weighted percentile surfaces.
LL/LH/HL/HH	Median-split co-location classes; HH = compounding. At the median split the table has one free number (HH); independence = 25 %.
Compounding intensity	Population-weighted mean of min(everyday pct, emergency pct); 1/3 independence, 1/2 coupled, 1/4 divergent.
Coverage grade	Severity ordering from the two clustering passes: covered / partial desert / desert.
Elasticity	Slope of log outcome on log population (space-for-time).
Wild cluster bootstrap	Country-clustered resampling giving reliable p-values with 24 clusters; the citable p for scaling claims.
p_{perm}	Permutation p-value reshuffling whole countries across macro-regions; the primary regional test.
TOST	Two one-sided equivalence tests; positive evidence a slope lies within a bound.
Specification curve	Headline regression re-estimated under every deprivation parameterisation.
Flip cells	Cells whose typology class changes under any deprivation variant.

S9 Per-city table

Table S11: The 67 functional urban areas, largest first. Grade: emergency-periphery coverage grade (Section 2.3); n_{em}: number of mapped emergency services (2 = both ED hospital and ambulance station); $\bar{D}_{\mathrm{ev}}$: population-weighted mean everyday deprivation level (0–1); $\bar{C}_{\mathrm{em}}$: mean emergency deprivation cost in multiples of the 15-min benchmark cost $g(15)$; G: population-weighted Gini per regime; ρ: Spearman coupling of the two percentile surfaces; HH: compounding population share at the median split.

City	CC	Region	Population	Grade	n_{em}	$\bar{D}_{\mathrm{ev}}$	$\bar{C}_{\mathrm{em}}$	G_{ev}	G_{em}	ρ	HH
Paris	FR	West	12,886,280	covered	2	0.135	0.73	0.625	0.497	0.50	0.334
Madrid	ES	South	7,601,394	covered	2	0.191	0.62	0.533	0.439	0.39	0.316
Barcelona	ES	South	5,945,709	covered	2	0.175	0.84	0.545	0.418	0.36	0.309
Milano	IT	South	5,030,024	covered	2	0.183	0.61	0.400	0.400	0.37	0.319
Berlin	DE	West	4,875,655	covered	2	0.158	0.62	0.558	0.438	0.38	0.317
Roma	IT	South	4,432,845	covered	2	0.287	1.14	0.424	0.503	0.48	0.342
Athina	EL	South	3,522,264	covered	1	0.103	0.73	0.609	0.500	0.28	0.274
Bruxelles	BE	West	3,521,497	covered	2	0.182	1.29	0.583	0.286	0.40	0.332
Napoli	IT	South	3,457,649	covered	2	0.326	0.83	0.305	0.384	0.15	0.270
Warszawa	PL	CEE	3,334,987	partial	2	0.225	1.99	0.589	0.590	0.49	0.347
Hamburg	DE	West	3,162,607	covered	2	0.260	0.58	0.543	0.437	0.40	0.321
Bucureşti	RO	CEE	3,140,502	desert	2	0.335	3.21	0.524	0.774	0.73	0.424
Lisboa	PT	South	3,132,904	covered	2	0.239	1.19	0.367	0.409	0.48	0.319
München	DE	West	3,054,507	covered	2	0.178	0.60	0.616	0.449	0.35	0.304
Budapest	HU	CEE	3,030,949	covered	2	0.183	1.05	0.538	0.507	0.48	0.327
Amsterdam	NL	West	2,829,893	covered	2	0.227	0.75	0.501	0.326	0.20	0.285
Stockholm	SE	North	2,549,795	partial	2	0.267	1.73	0.460	0.513	0.65	0.389
Köln	DE	West	2,438,338	covered	2	0.204	0.57	0.520	0.398	0.36	0.312
Praha	CZ	CEE	2,247,160	covered	2	0.282	1.11	0.515	0.475	0.46	0.354
København	DK	North	2,180,998	covered	2	0.254	0.94	0.401	0.401	0.30	0.307
Sofia	BG	CEE	1,647,936	partial	2	0.285	1.44	0.511	0.600	0.73	0.408
Helsinki	FI	North	1,642,879	covered	1	0.379	1.36	0.341	0.426	0.26	0.270
Oslo	NO	North	1,550,450	partial	2	0.209	2.06	0.553	0.467	0.39	0.297
Porto	PT	South	1,531,690	covered	2	0.324	0.85	0.272	0.395	0.42	0.321
Kraków	PL	CEE	1,350,286	covered	2	0.308	1.50	0.523	0.519	0.58	0.396
Katowice	PL	CEE	1,341,890	covered	2	0.168	1.02	0.489	0.279	0.10	0.267
Zagreb	HR	CEE	1,293,698	partial	2	0.343	2.07	0.499	0.583	0.70	0.395
Antwerpen	BE	West	1,219,366	covered	2	0.289	0.77	0.523	0.406	0.29	0.311
Rīga	LV	CEE	1,110,303	desert	2	0.362	3.37	0.502	0.682	0.55	0.374
Bilbao	ES	South	1,053,444	covered	2	0.160	1.06	0.570	0.433	0.42	0.324
Łódź	PL	CEE	1,045,886	covered	2	0.255	1.60	0.534	0.491	0.33	0.310
Göteborg	SE	North	1,042,720	partial	2	0.427	2.44	0.335	0.446	0.31	0.298
Vilnius	LT	CEE	1,010,376	desert	2	0.462	5.39	0.411	0.739	0.55	0.358
Palermo	IT	South	1,006,055	partial	2	0.317	3.64	0.383	0.292	0.27	0.289
Alicante	ES	South	996,867	covered	2	0.311	1.80	0.530	0.277	0.15	0.269
Wrocław	PL	CEE	939,076	covered	2	0.199	1.47	0.609	0.362	0.41	0.341
Bratislava	SK	CEE	931,613	covered	2	0.296	1.78	0.522	0.430	0.39	0.336
Ljubljana	SI	CEE	919,303	desert	2	0.447	4.96	0.387	0.481	0.48	0.341
Palma de Mallorca	ES	South	818,202	covered	2	0.251	1.39	0.503	0.453	0.14	0.294
Brno	CZ	CEE	745,397	covered	2	0.286	1.14	0.561	0.457	0.60	0.369
Tallinn	EE	CEE	731,883	desert	2	0.349	2.80	0.456	0.713	0.45	0.326
Grenoble	FR	West	688,170	covered	2	0.211	1.11	0.583	0.444	0.62	0.379
Luxembourg	LU	West	595,677	partial	2	0.312	1.64	0.469	0.407	0.30	0.301
Toulon	FR	West	540,279	covered	2	0.315	0.91	0.431	0.458	0.41	0.326
Aarhus	DK	North	525,545	covered	2	0.336	1.81	0.394	0.438	0.38	0.310
Chemnitz	DE	West	481,607	covered	2	0.223	0.69	0.558	0.363	0.36	0.319
Turku	FI	North	353,992	partial	2	0.409	1.63	0.385	0.565	0.67	0.360
Stavanger	NO	North	331,780	partial	2	0.252	2.10	0.498	0.492	0.28	0.299
Aalborg	DK	North	309,890	partial	2	0.405	1.87	0.419	0.541	0.47	0.342
Caserta	IT	South	298,402	covered	2	0.283	1.33	0.298	0.228	0.18	0.272
Vicenza	IT	South	280,054	covered	2	0.275	1.08	0.391	0.411	0.47	0.344
Helsingborg	SE	North	253,346	covered	2	0.420	0.90	0.284	0.387	0.51	0.366

Table S11 continued

City	CC	Region	Population	Grade	n_{em}	$\bar{D}_{\mathrm{ev}}$	$\bar{C}_{\mathrm{em}}$	G_{ev}	G_{em}	ρ	HH
Jaén	ES	South	249,477	covered	2	0.291	1.24	0.543	0.385	-0.02	0.233
Szeged	HU	CEE	246,224	covered	1	0.281	0.82	0.533	0.568	0.58	0.392
Apeldoorn	NL	West	242,638	covered	2	0.326	1.02	0.439	0.341	0.33	0.333
Cosenza	IT	South	239,305	covered	2	0.505	1.23	0.407	0.500	0.70	0.390
Brăila	RO	CEE	235,919	partial	1	0.422	1.29	0.434	0.692	0.63	0.359
Amersfoort	NL	West	234,650	covered	2	0.219	0.64	0.458	0.350	0.26	0.282
Žilina	SK	CEE	217,546	covered	1	0.431	1.46	0.424	0.443	0.62	0.365
Västerås	SE	North	193,510	covered	2	0.454	1.13	0.327	0.478	0.62	0.387
Norrköping	SE	North	193,317	partial	1	0.445	1.88	0.423	0.560	0.55	0.353
Lahti	FI	North	191,532	covered	1	0.495	2.14	0.353	0.444	0.50	0.345
Landshut	DE	West	168,617	covered	2	0.364	0.68	0.492	0.438	0.49	0.361
Sittard-Geleen	NL	West	165,544	covered	2	0.440	0.83	0.316	0.273	0.31	0.287
Arezzo	IT	South	146,788	covered	2	0.381	1.02	0.412	0.446	0.45	0.342
Łomża	PL	CEE	117,255	partial	2	0.484	1.01	0.417	0.621	0.74	0.430
Talavera de la Reina	ES	South	101,050	covered	1	0.378	0.89	0.522	0.454	0.66	0.415

S10 Per-city compounding maps

The median-split co-location typology map of each of the 67 cities, ordered by population. Legends carry population-weighted class shares; the maps are area-weighted, so coloured area and quoted share will not match by eye. Read for spatial pattern only: roughly 24 % of population flips class between routing engines (E.1), so no single cell's class is a finding.

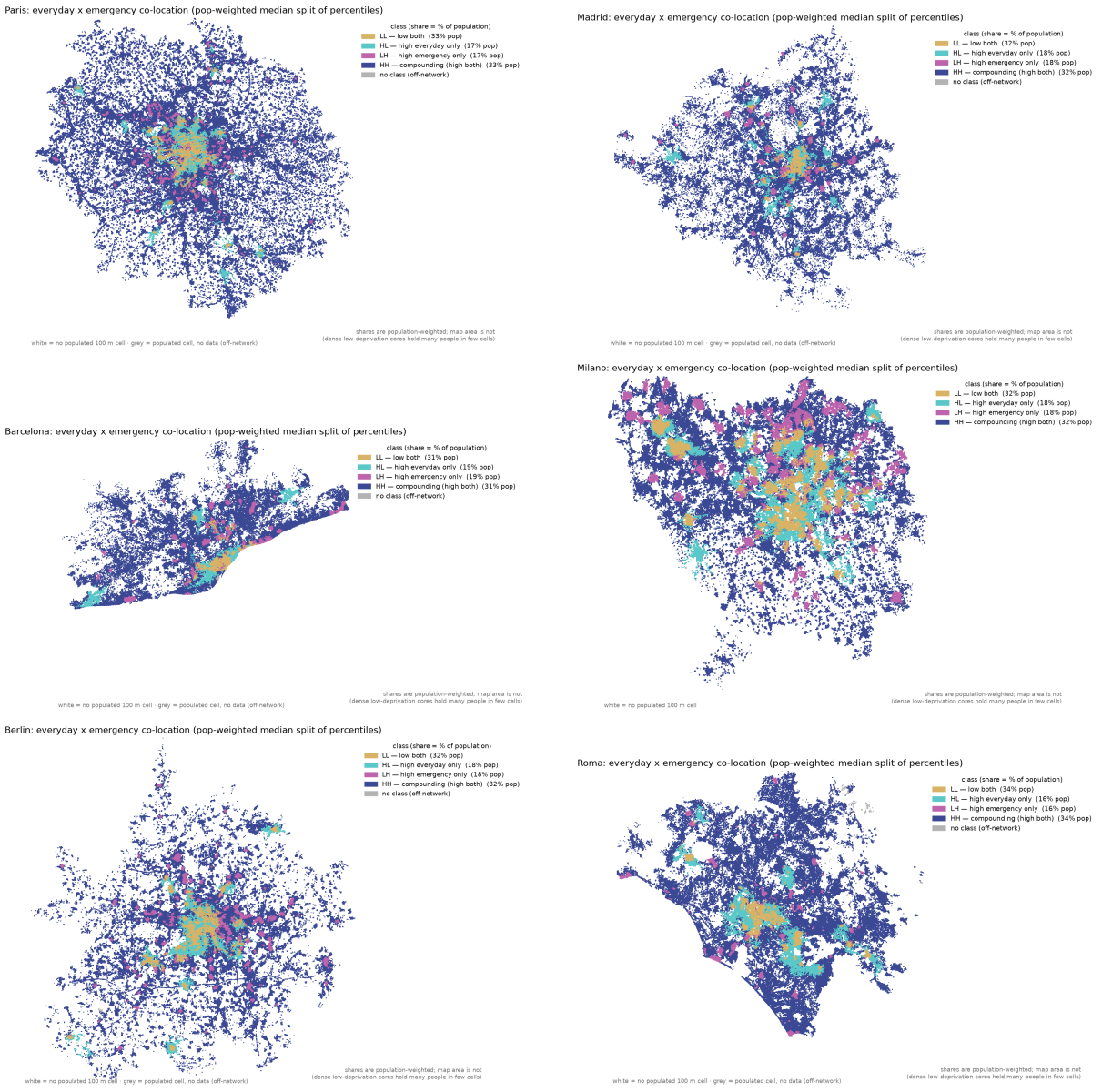


Figure S5: Per-city compounding maps (median-split co-location typology), cities 1–6 by population: Paris (FR), Madrid (ES), Barcelona (ES), Milano (IT), Berlin (DE), Roma (IT). Read for spatial pattern only; per-cell classes carry a $\sim$24 % routing-engine flip share (E.1).

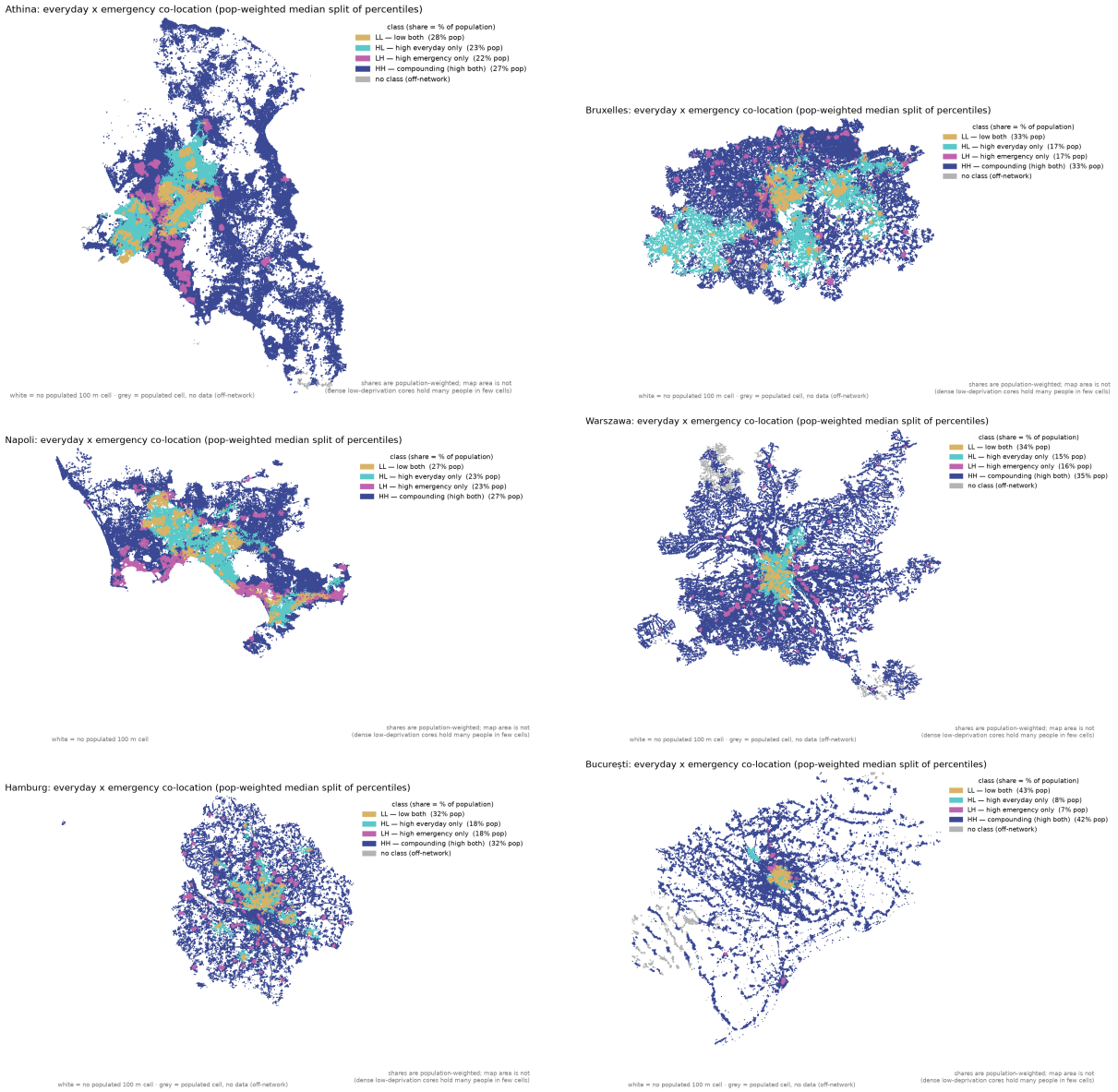


Figure S6: Per-city compounding maps (median-split co-location typology), cities 7–12 by population: Athina (EL), Bruxelles (BE), Napoli (IT), Warszawa (PL), Hamburg (DE), Bucureşti (RO). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

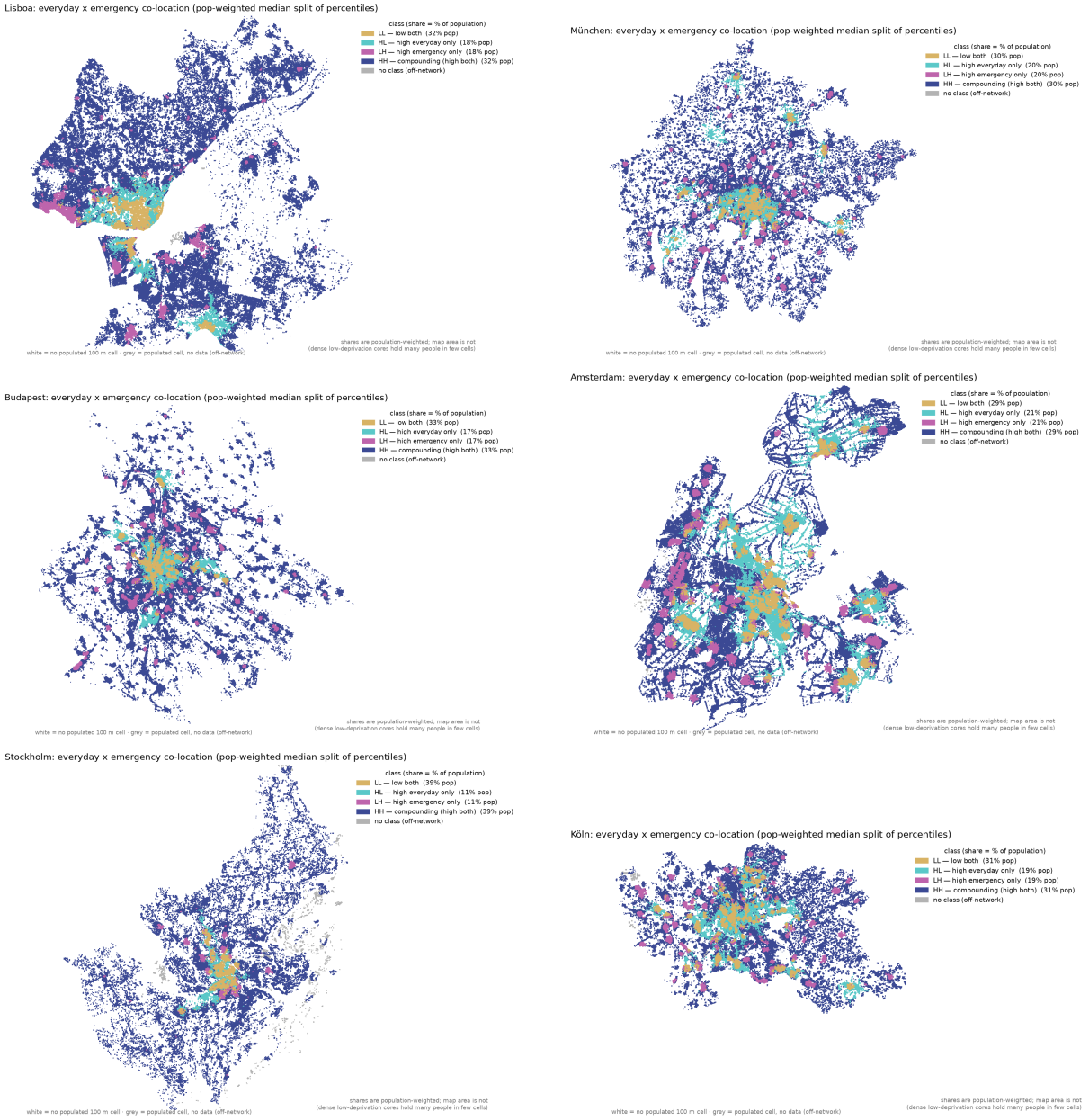


Figure S7: Per-city compounding maps (median-split co-location typology), cities 13–18 by population: Lisboa (PT), München (DE), Budapest (HU), Amsterdam (NL), Stockholm (SE), Köln (DE). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

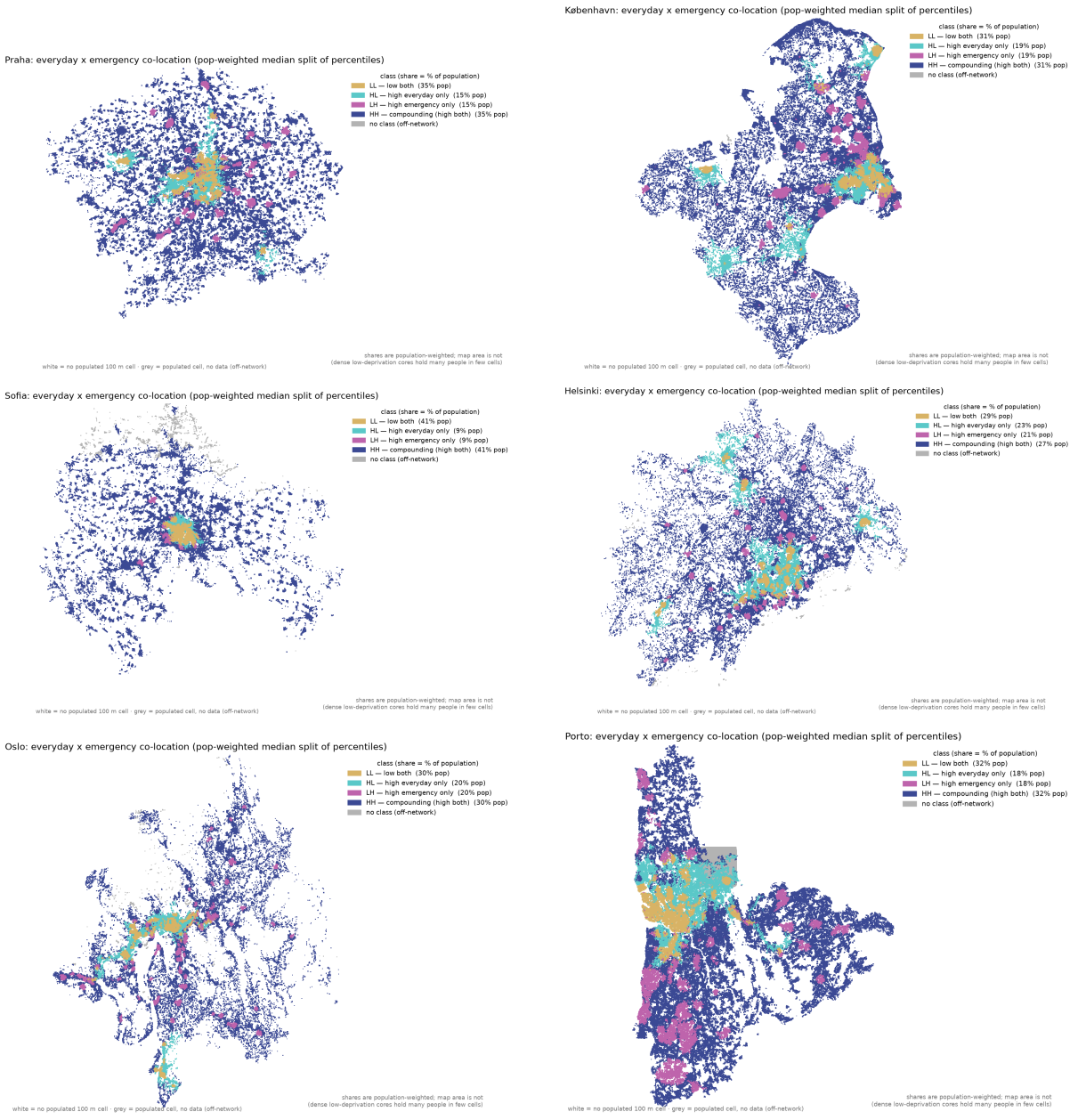


Figure S8: Per-city compounding maps (median-split co-location typology), cities 19–24 by population: Praha (CZ), København (DK), Sofia (BG), Helsinki (FI), Oslo (NO), Porto (PT). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

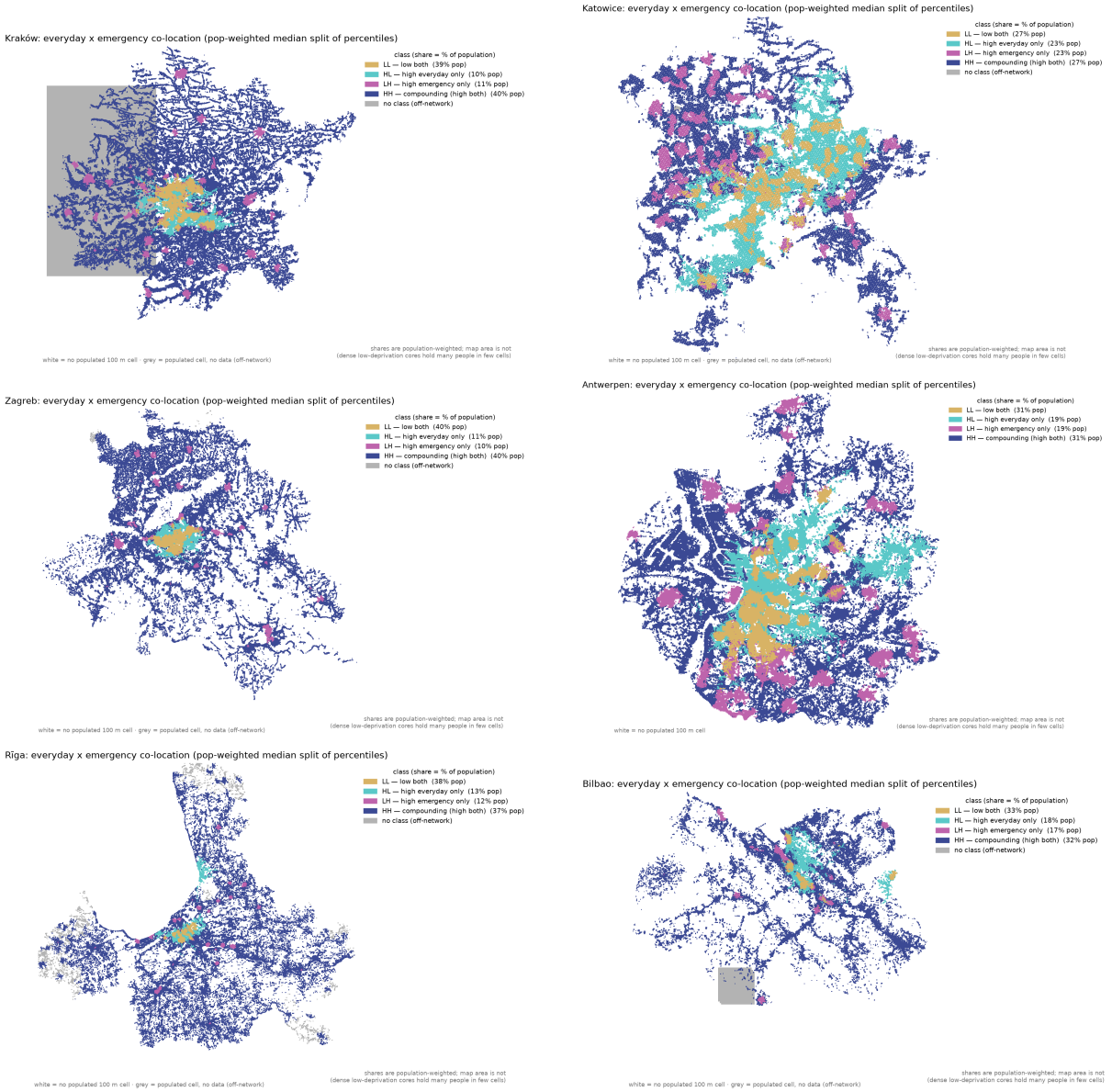


Figure S9: Per-city compounding maps (median-split co-location typology), cities 25–30 by population: Kraków (PL), Katowice (PL), Zagreb (HR), Antwerpen (BE), Rīga (LV), Bilbao (ES). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

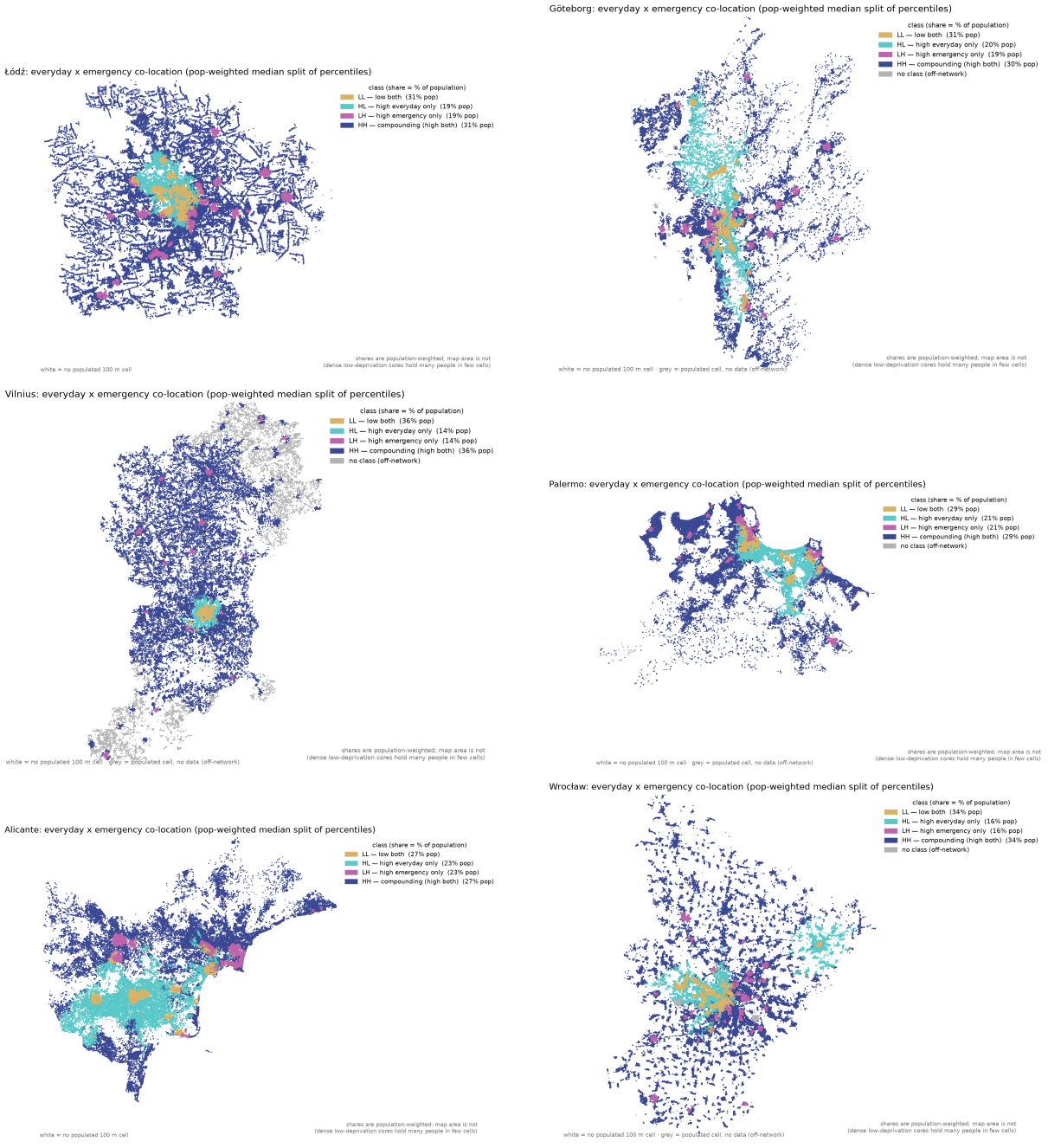


Figure S10: Per-city compounding maps (median-split co-location typology), cities 31–36 by population: Łódź (PL), Göteborg (SE), Vilnius (LT), Palermo (IT), Alicante (ES), Wrocław (PL). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).



Figure S11: Per-city compounding maps (median-split co-location typology), cities 37–42 by population: Bratislava (SK), Ljubljana (SI), Palma de Mallorca (ES), Brno (CZ), Tallinn (EE), Grenoble (FR). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

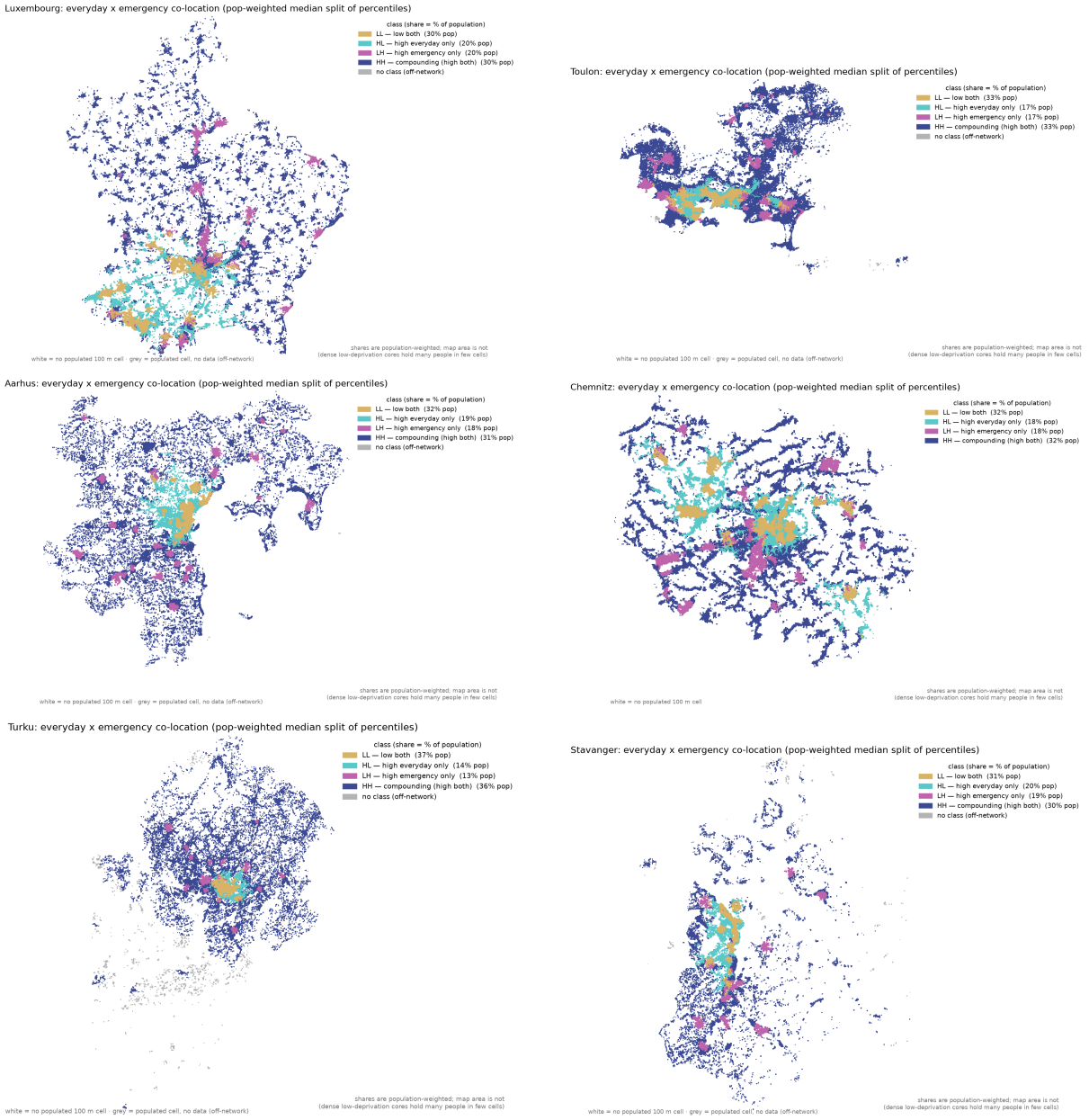


Figure S12: Per-city compounding maps (median-split co-location typology), cities 43–48 by population: Luxembourg (LU), Toulon (FR), Aarhus (DK), Chemnitz (DE), Turku (FI), Stavanger (NO). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

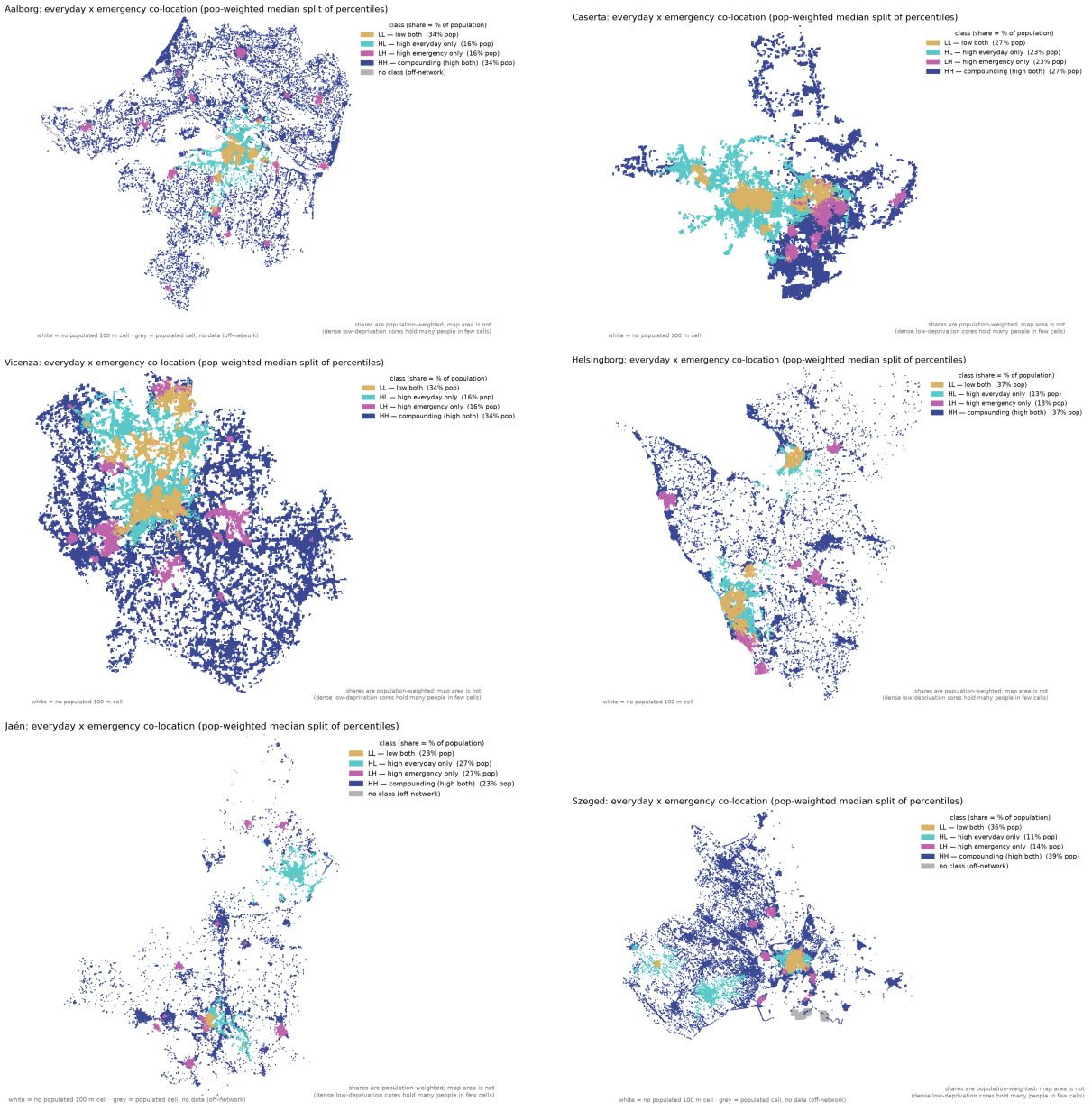


Figure S13: Per-city compounding maps (median-split co-location typology), cities 49–54 by population: Aalborg (DK), Caserta (IT), Vicenza (IT), Helsingborg (SE), Jaén (ES), Szeged (HU). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

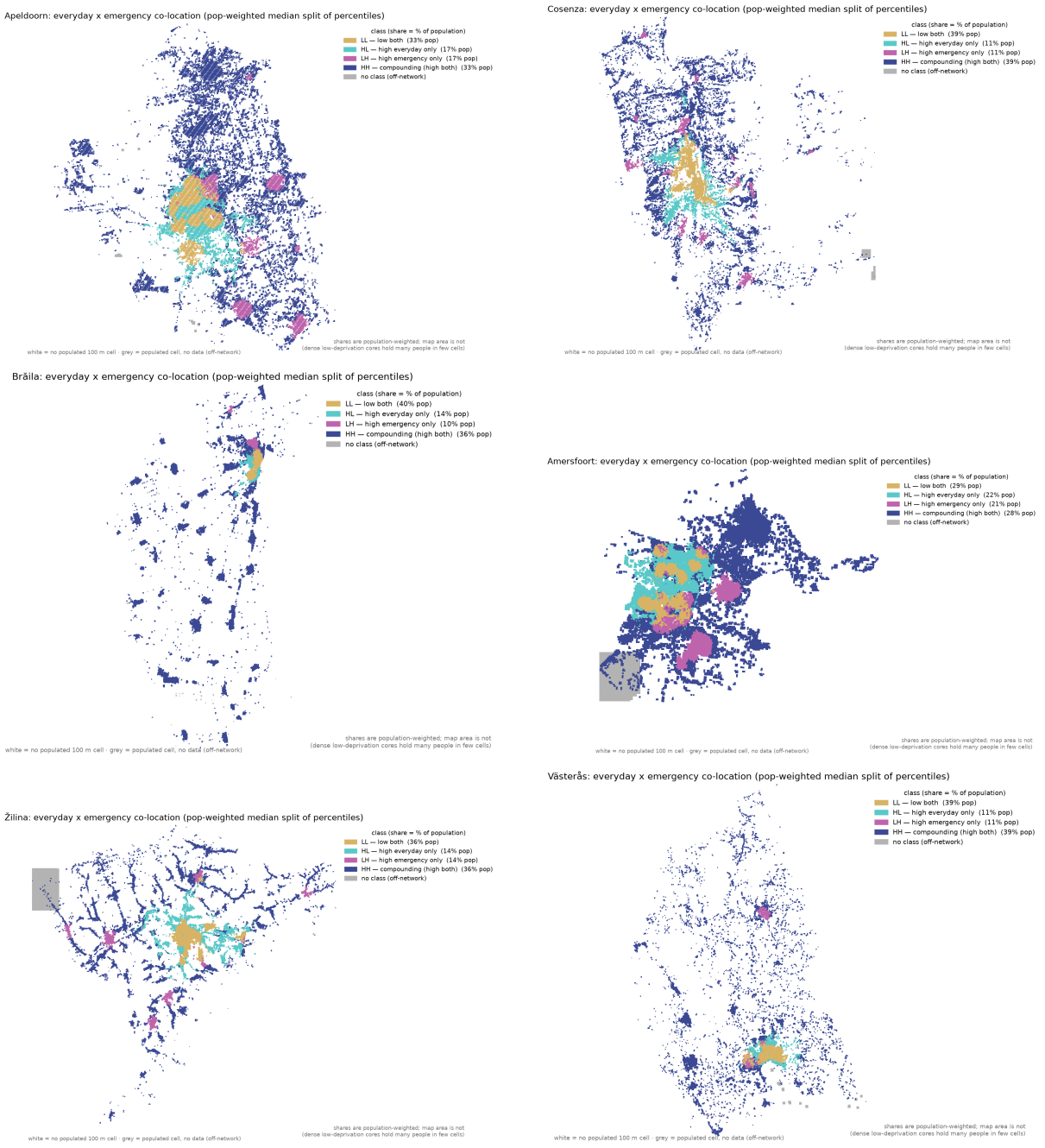


Figure S14: Per-city compounding maps (median-split co-location typology), cities 55–60 by population: Apeldoorn (NL), Cosenza (IT), Brăila (RO), Amersfoort (NL), Žilina (SK), Västerås (SE). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

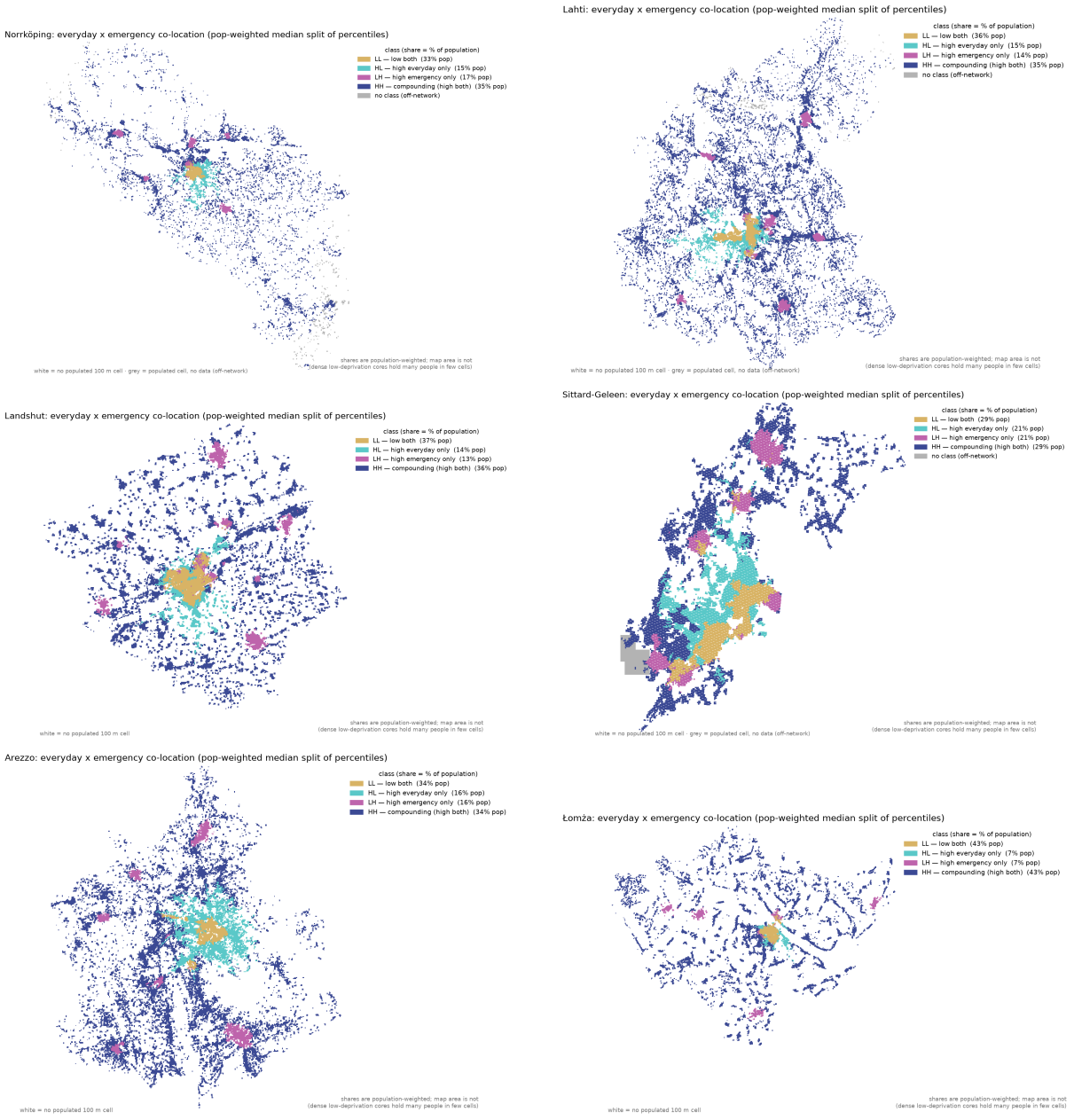


Figure S15: Per-city compounding maps (median-split co-location typology), cities 61–66 by population: Norrköping (SE), Lahti (FI), Landshut (DE), Sittard-Geleen (NL), Arezzo (IT), Łomża (PL). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).

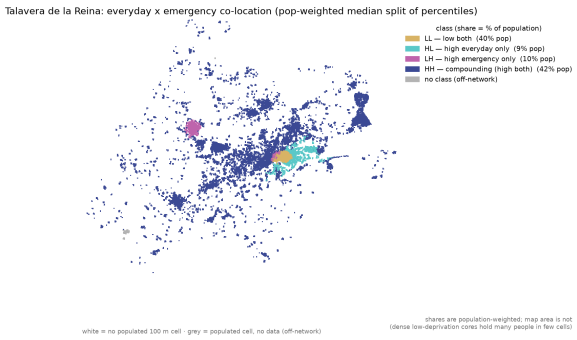


Figure S16: Per-city compounding maps (median-split co-location typology), cities 67–67 by population: Talavera de la Reina (ES). Read for spatial pattern only; per-cell classes carry a ∼24 % routing-engine flip share (E.1).